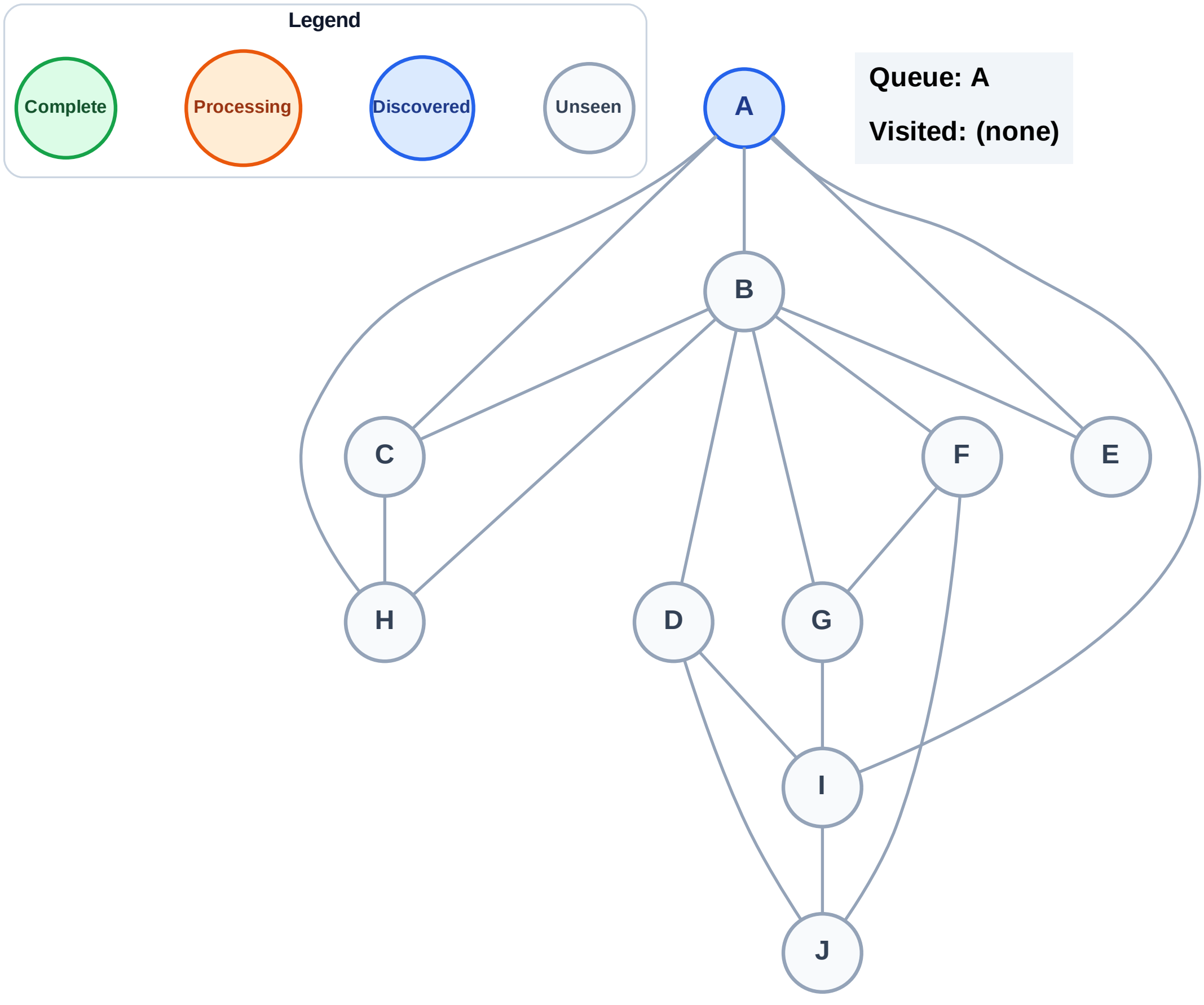
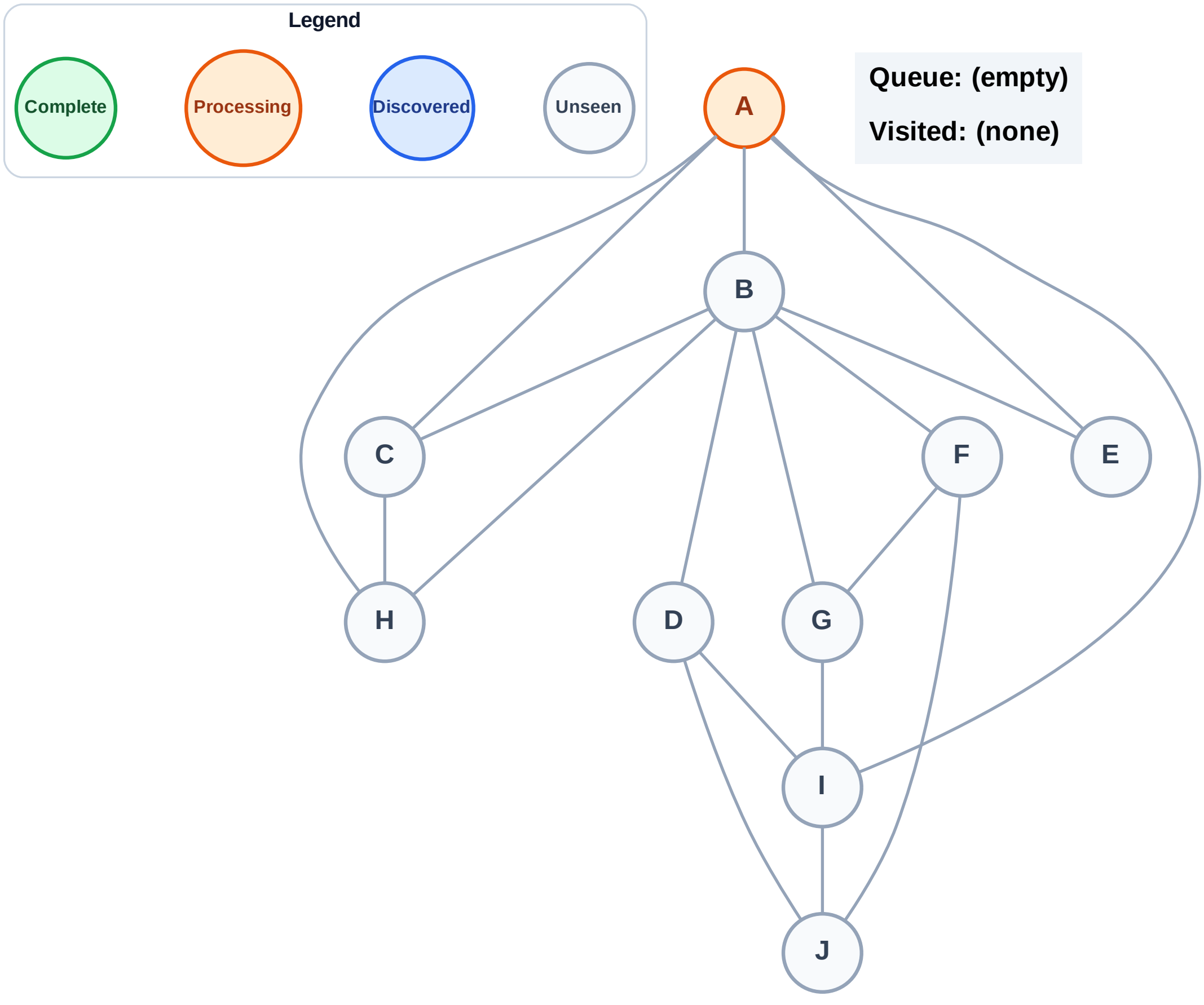
BFS Traversal

Step 1: Start at A



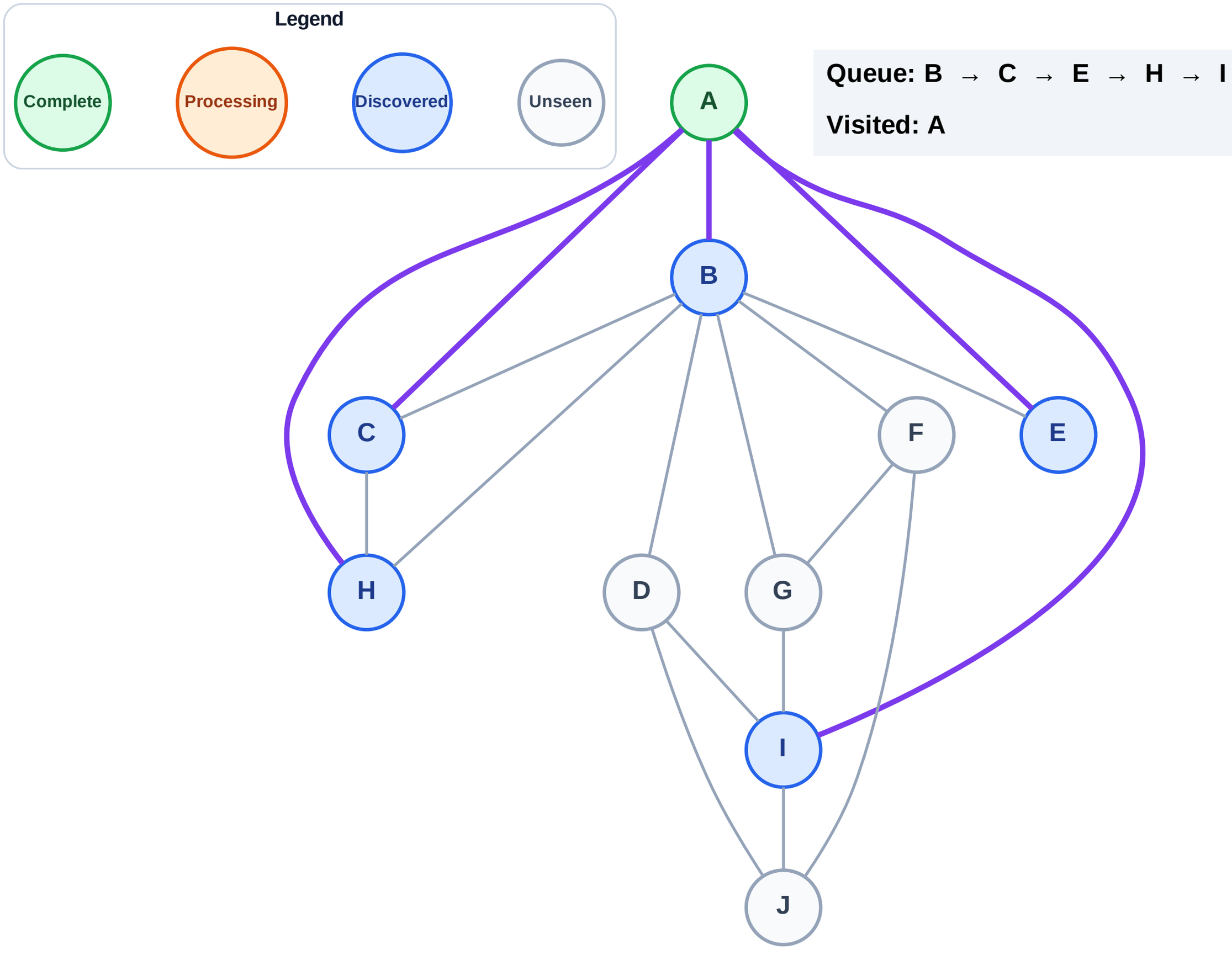
BFS Traversal

Step 2: Process node A



BFS Traversal

Step 3: Discover B, C, E, H, I from A

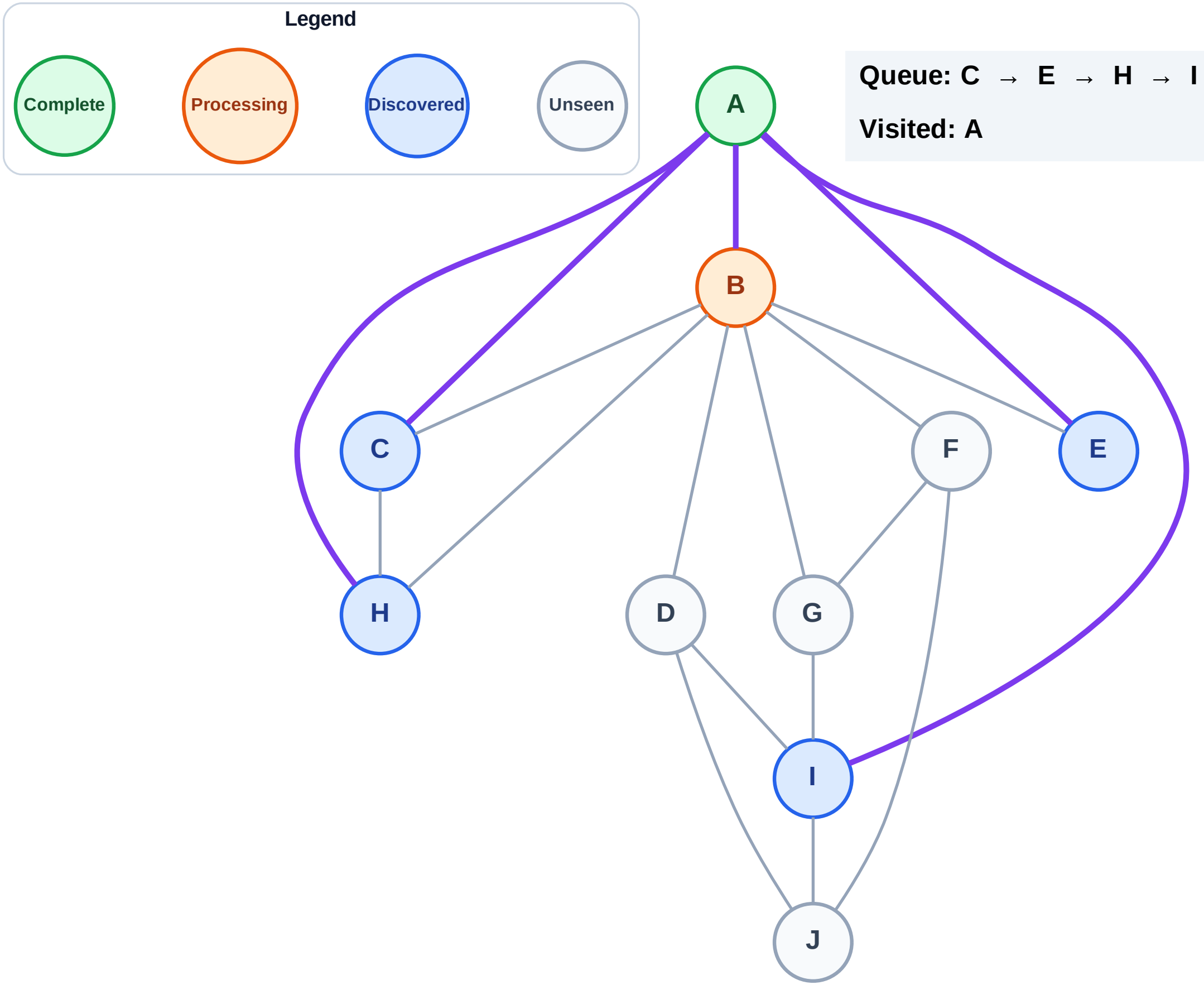


Queue: B → C → E → H → I

Visited: A

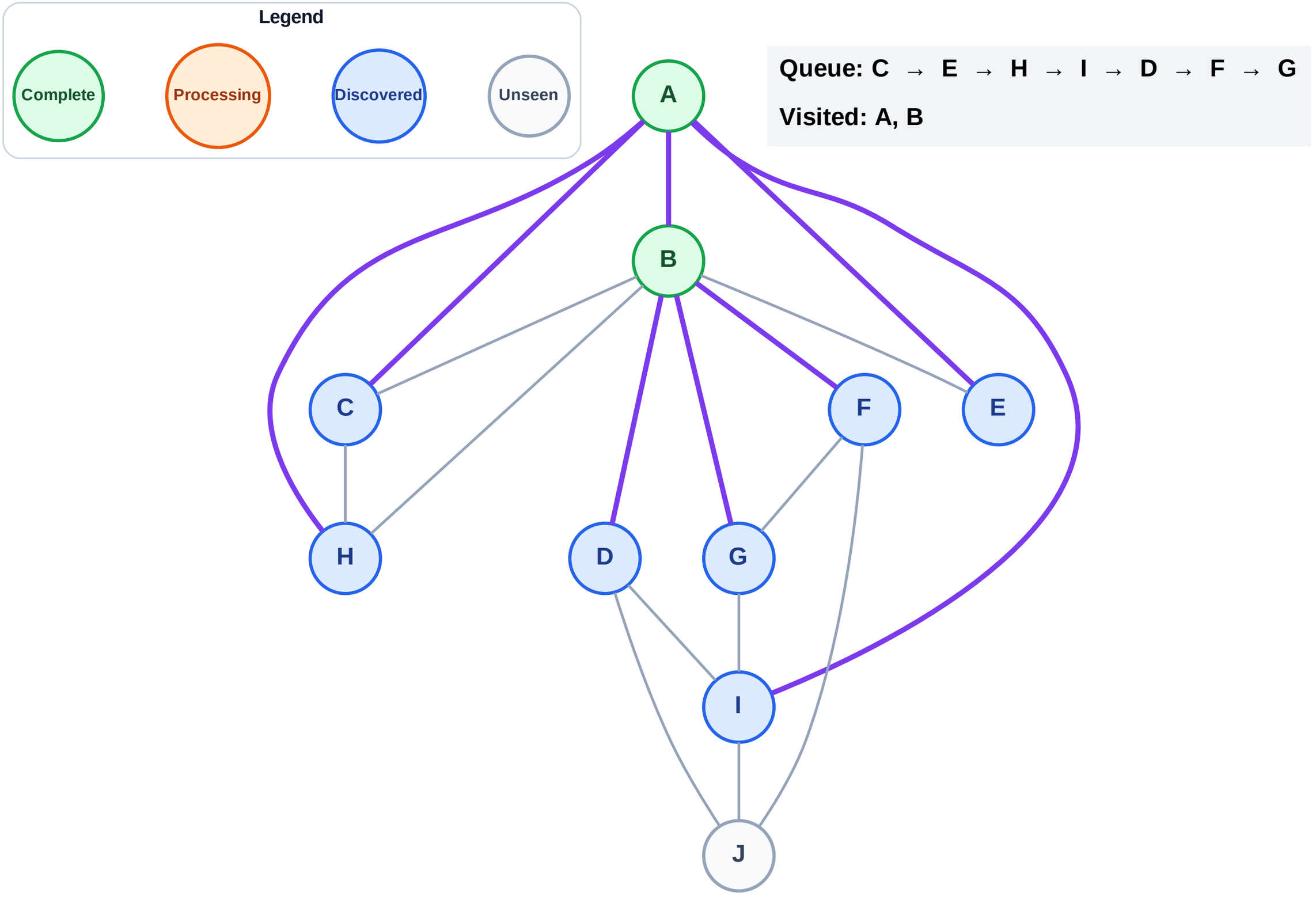
BFS Traversal

Step 4: Process node B



BFS Traversal

Step 5: Discover D, F, G from B

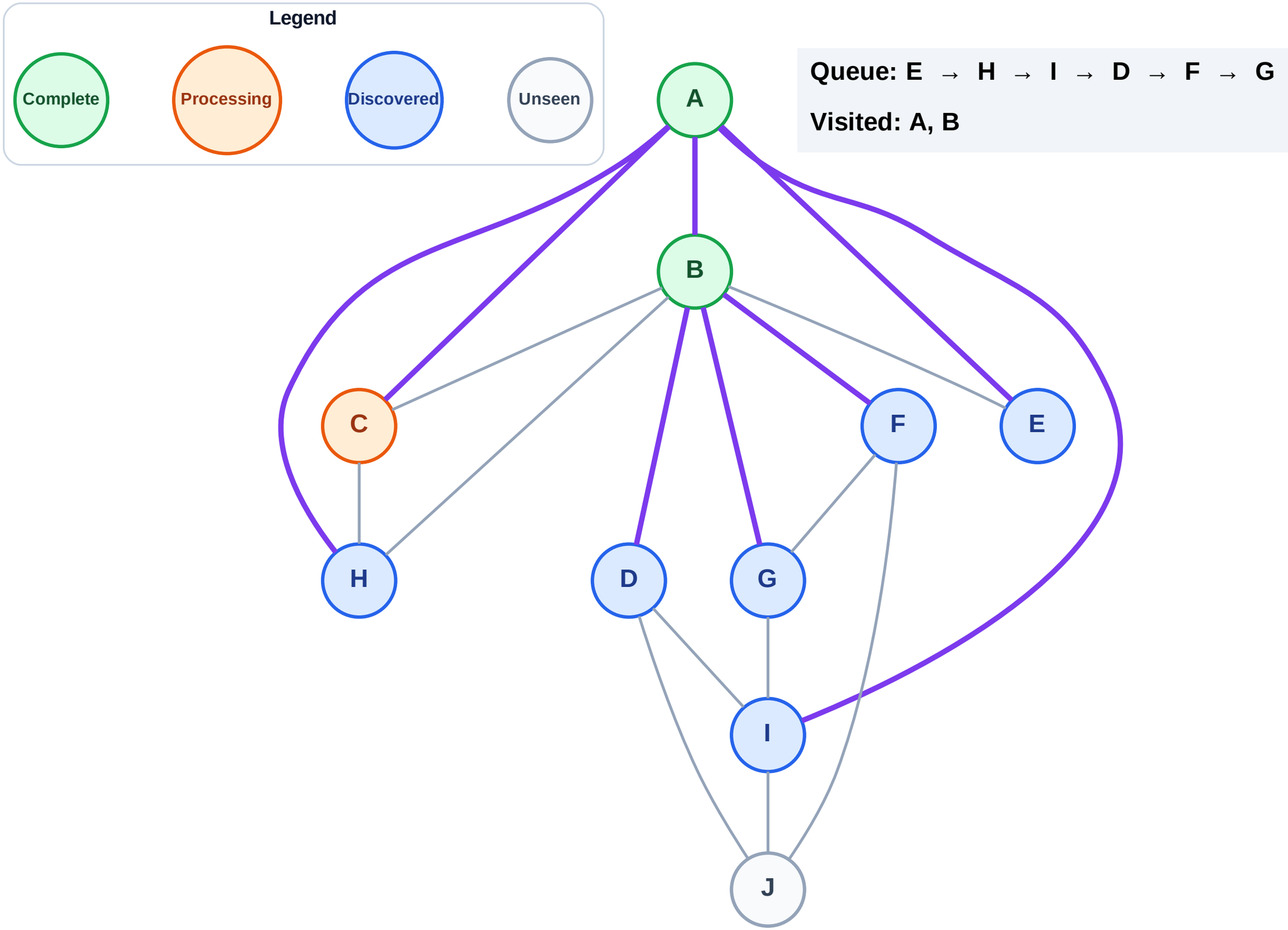


Queue: C → E → H → I → D → F → G

Visited: A, B

BFS Traversal

Step 6: Process node C

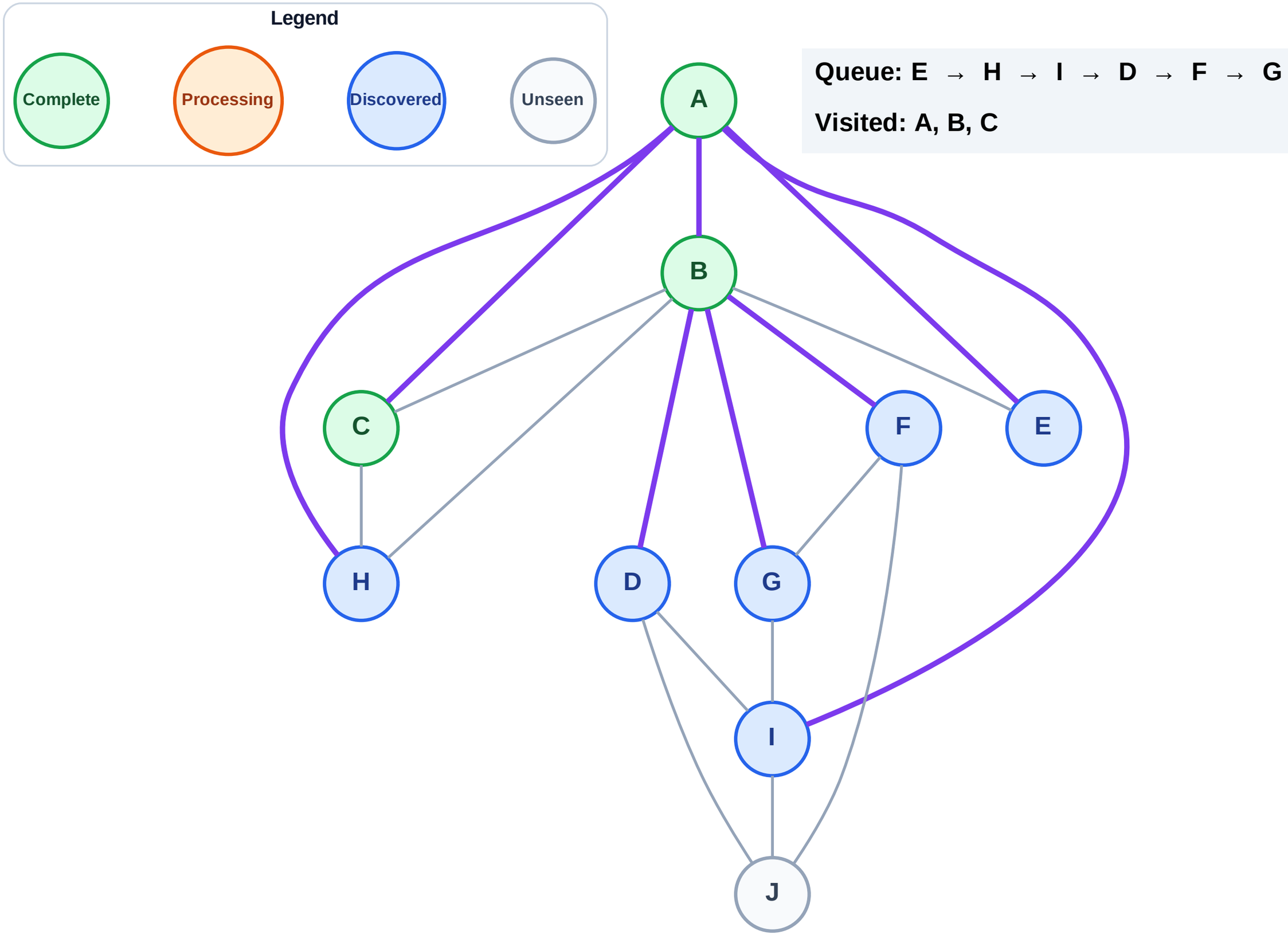


Queue: E → H → I → D → F → G

Visited: A, B

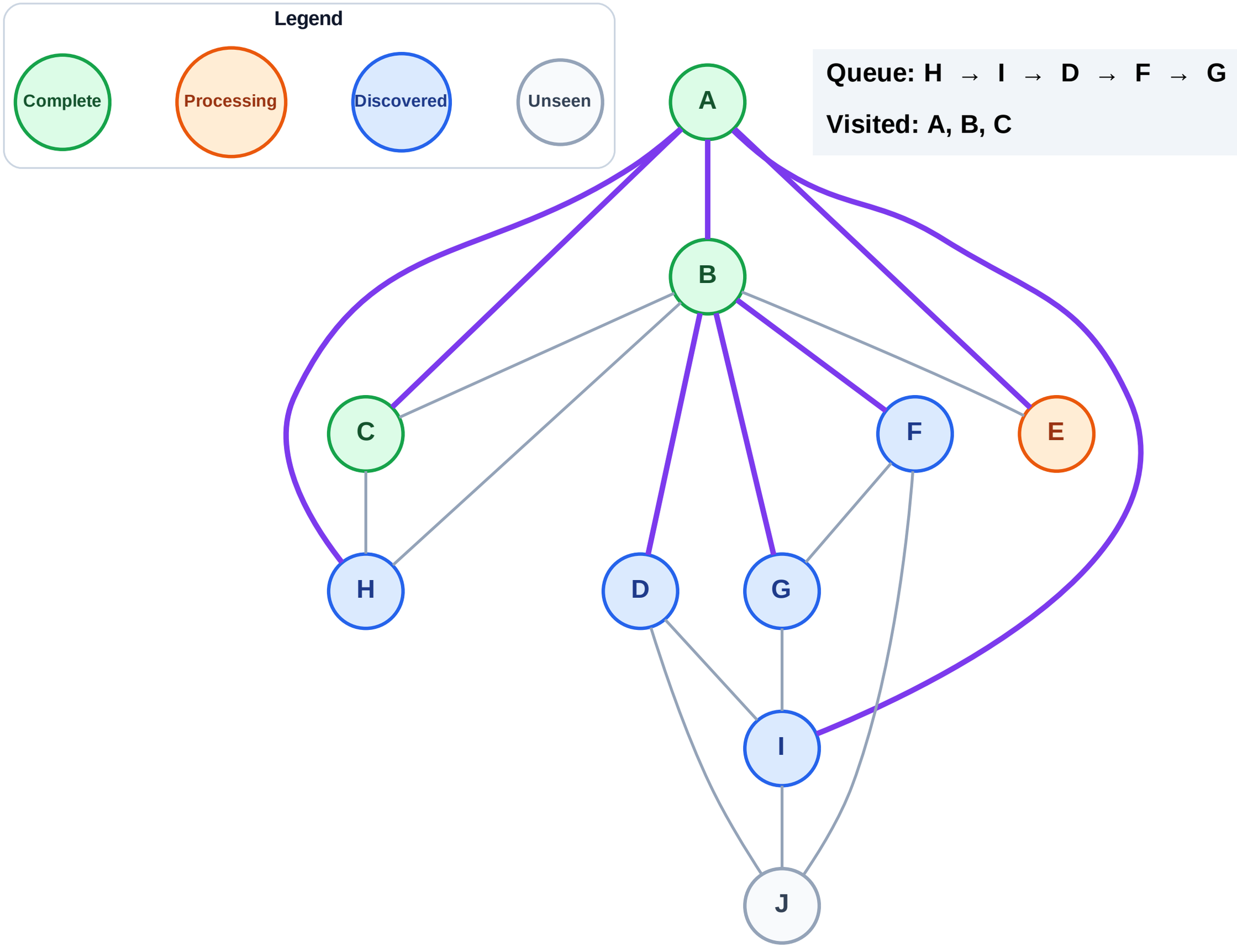
BFS Traversal

Step 7: No new neighbors from C



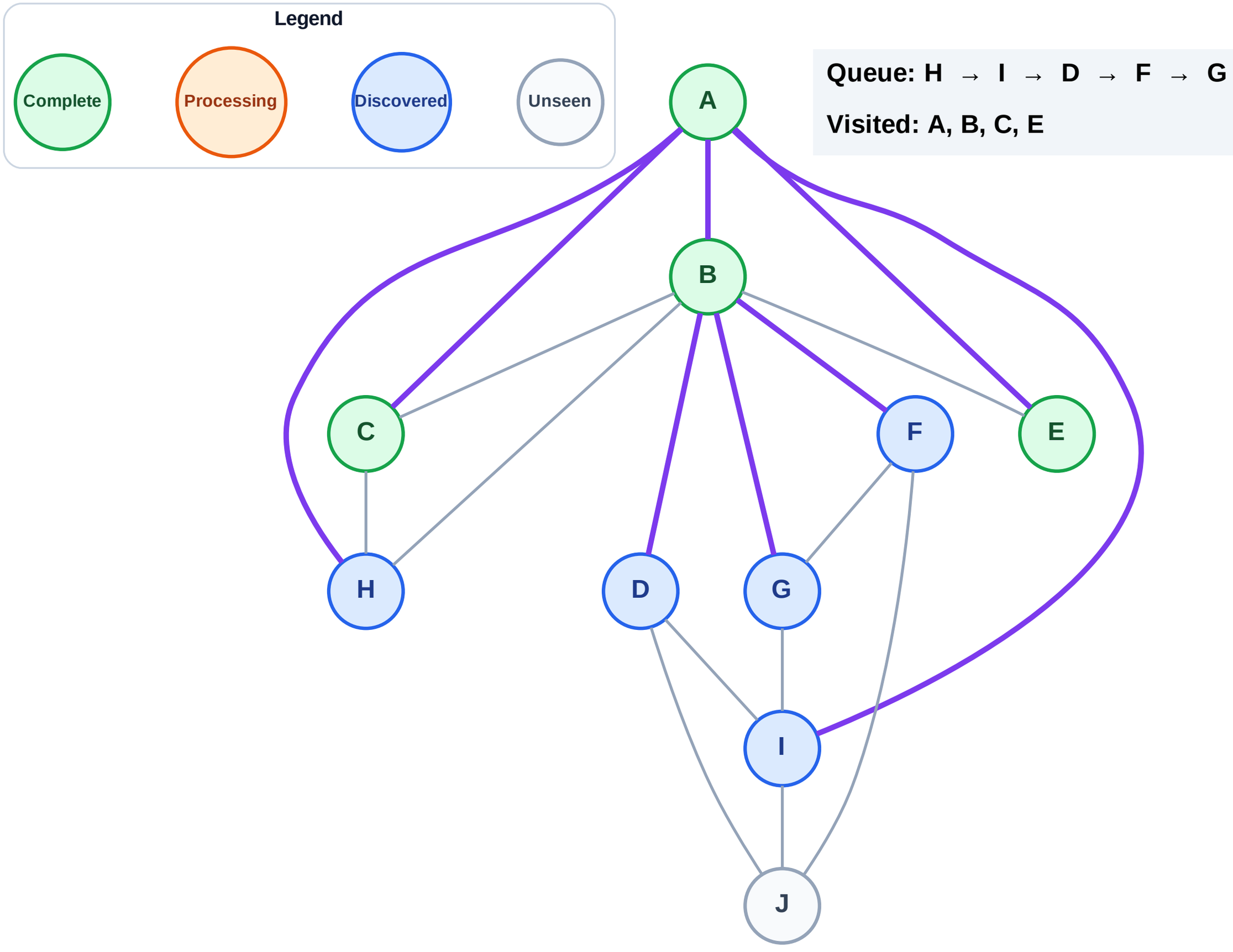
BFS Traversal

Step 8: Process node E



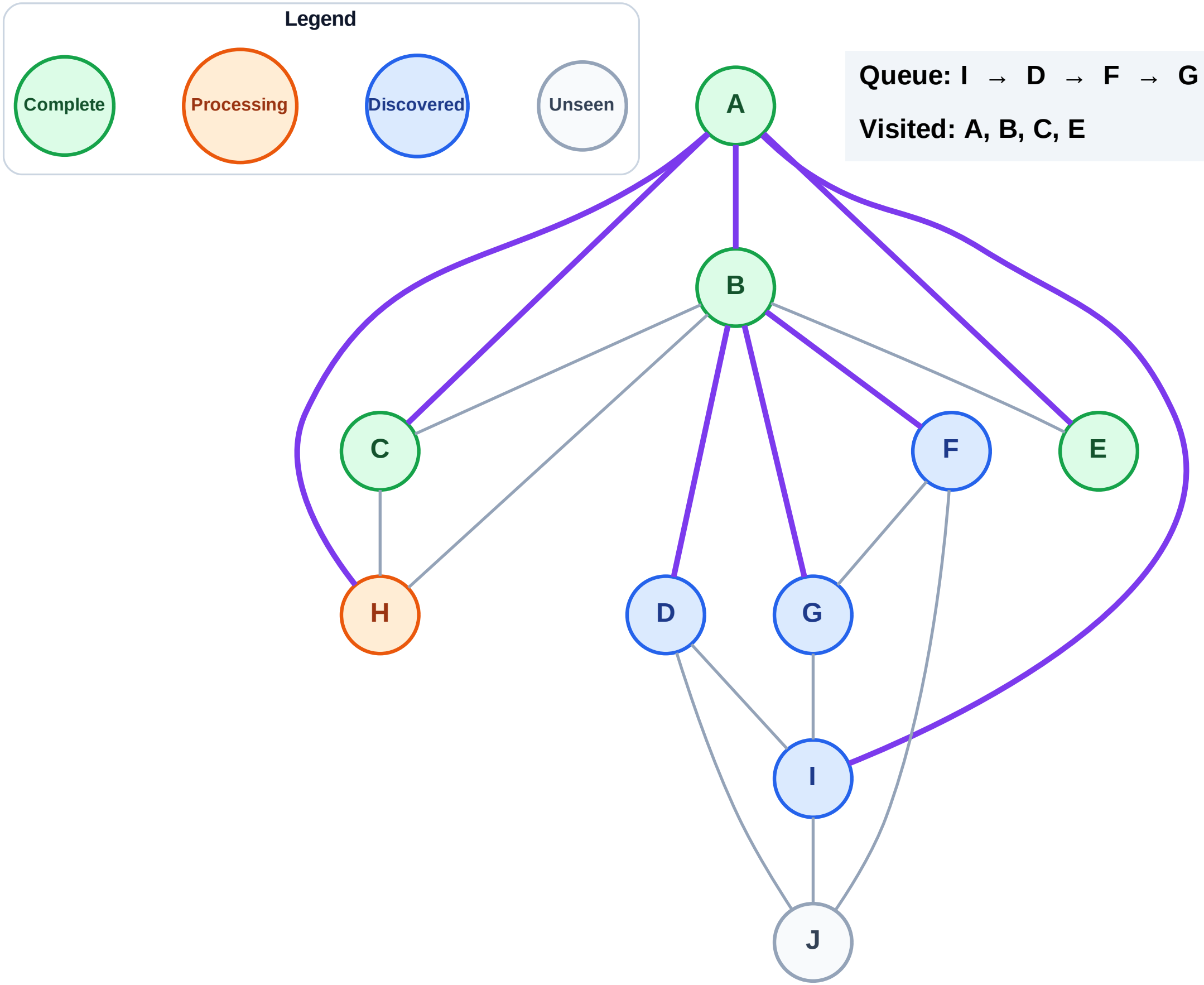
BFS Traversal

Step 9: No new neighbors from E



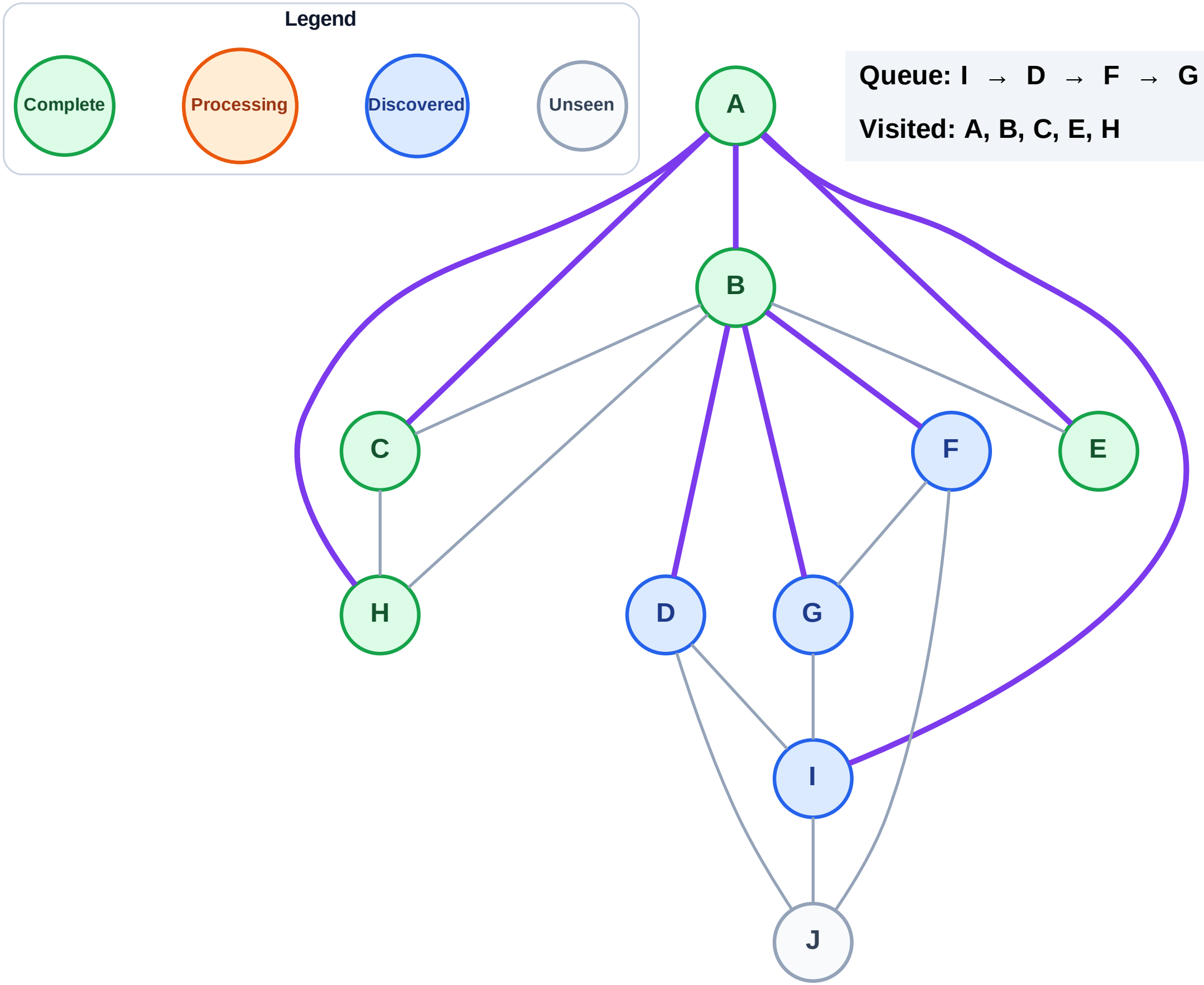
BFS Traversal

Step 10: Process node H



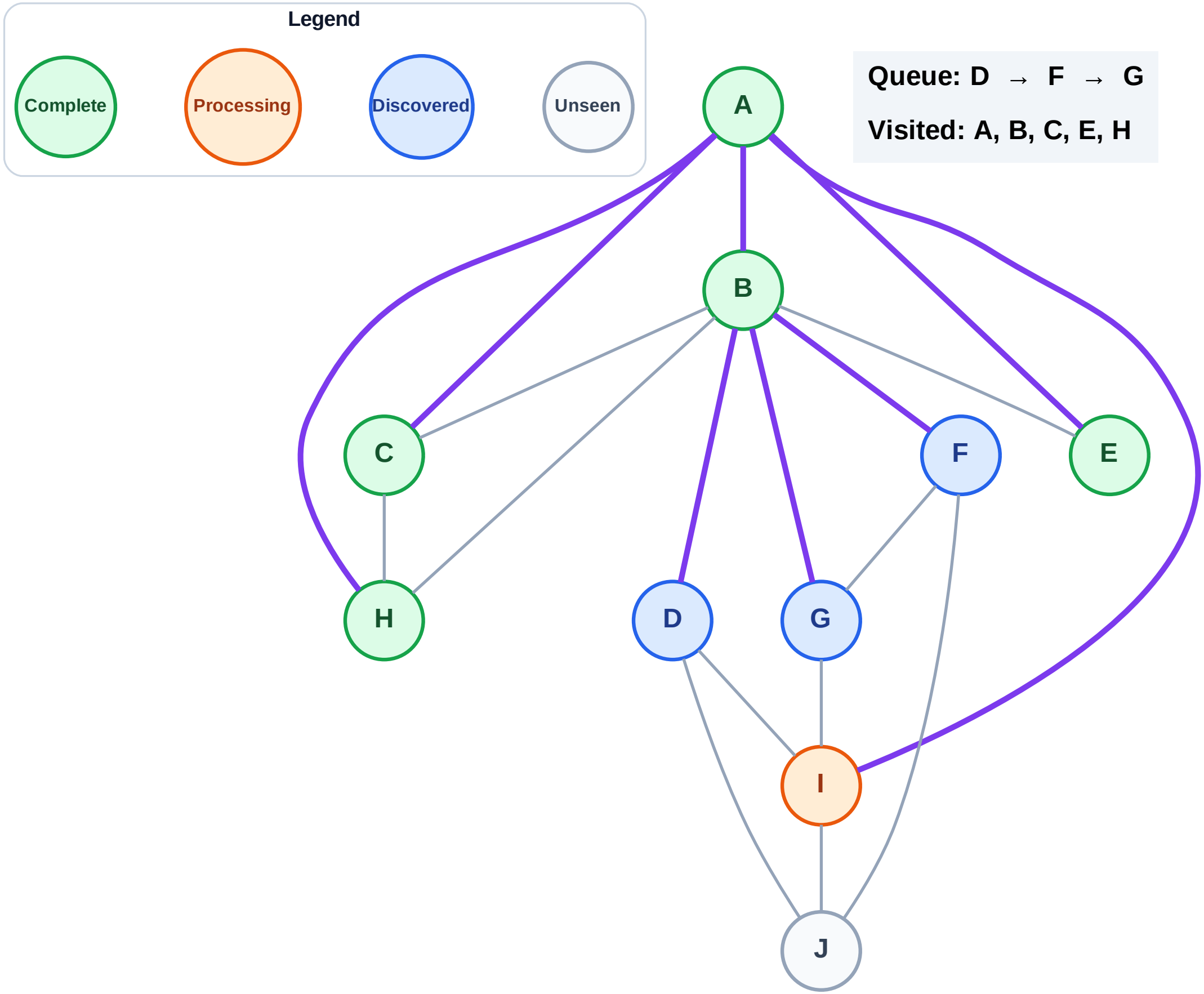
BFS Traversal

Step 11: No new neighbors from H



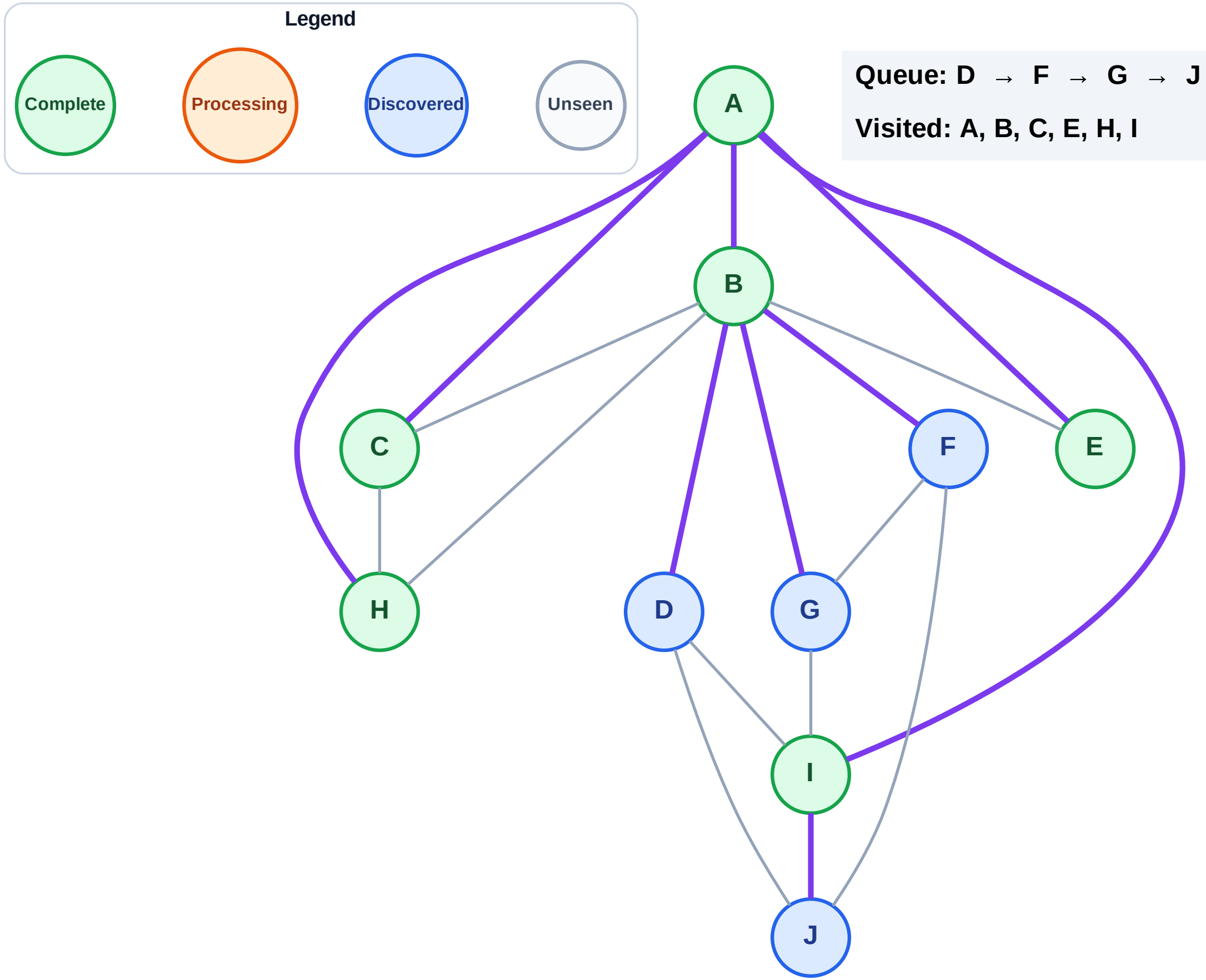
BFS Traversal

Step 12: Process node I



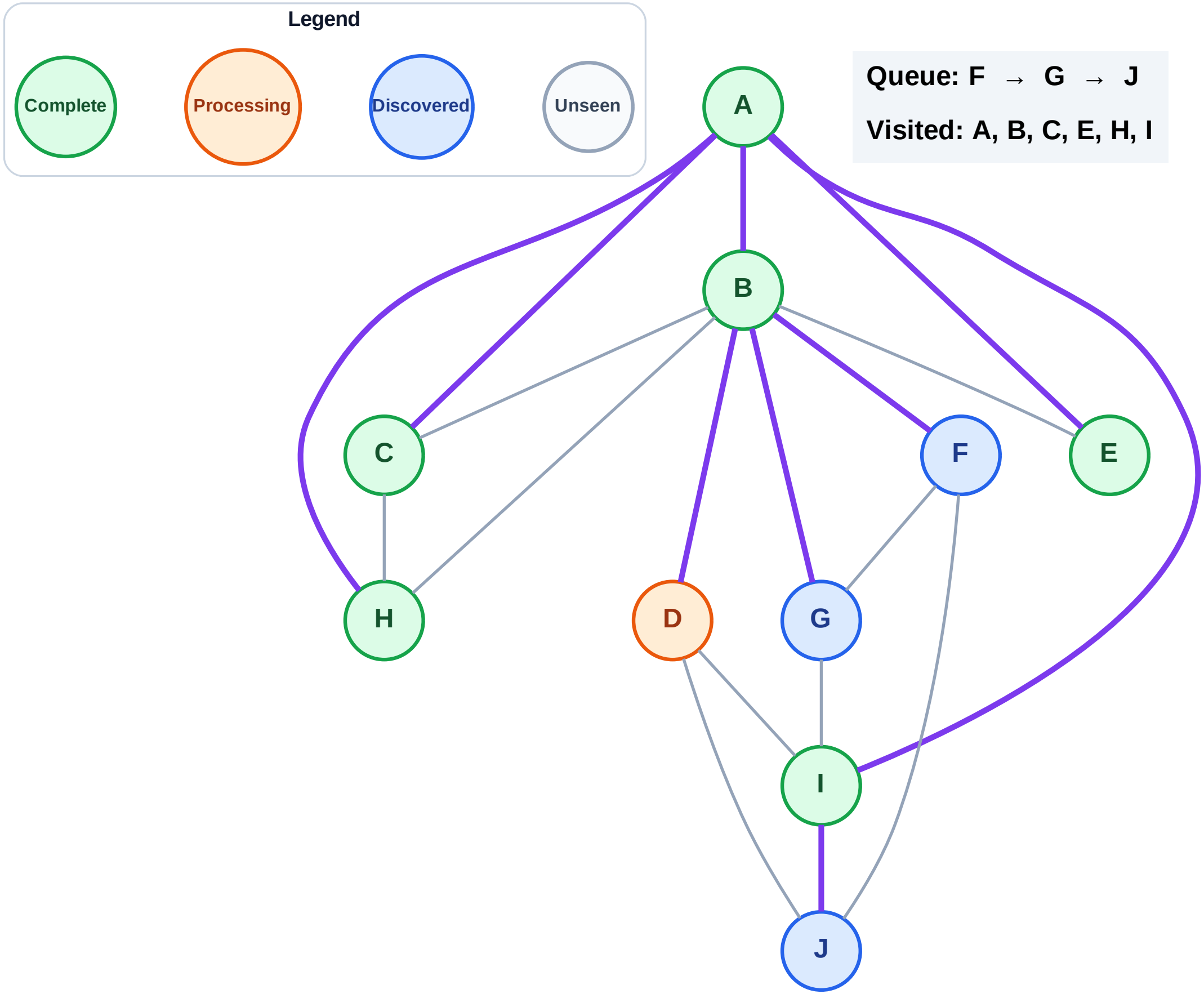
BFS Traversal

Step 13: Discover J from I



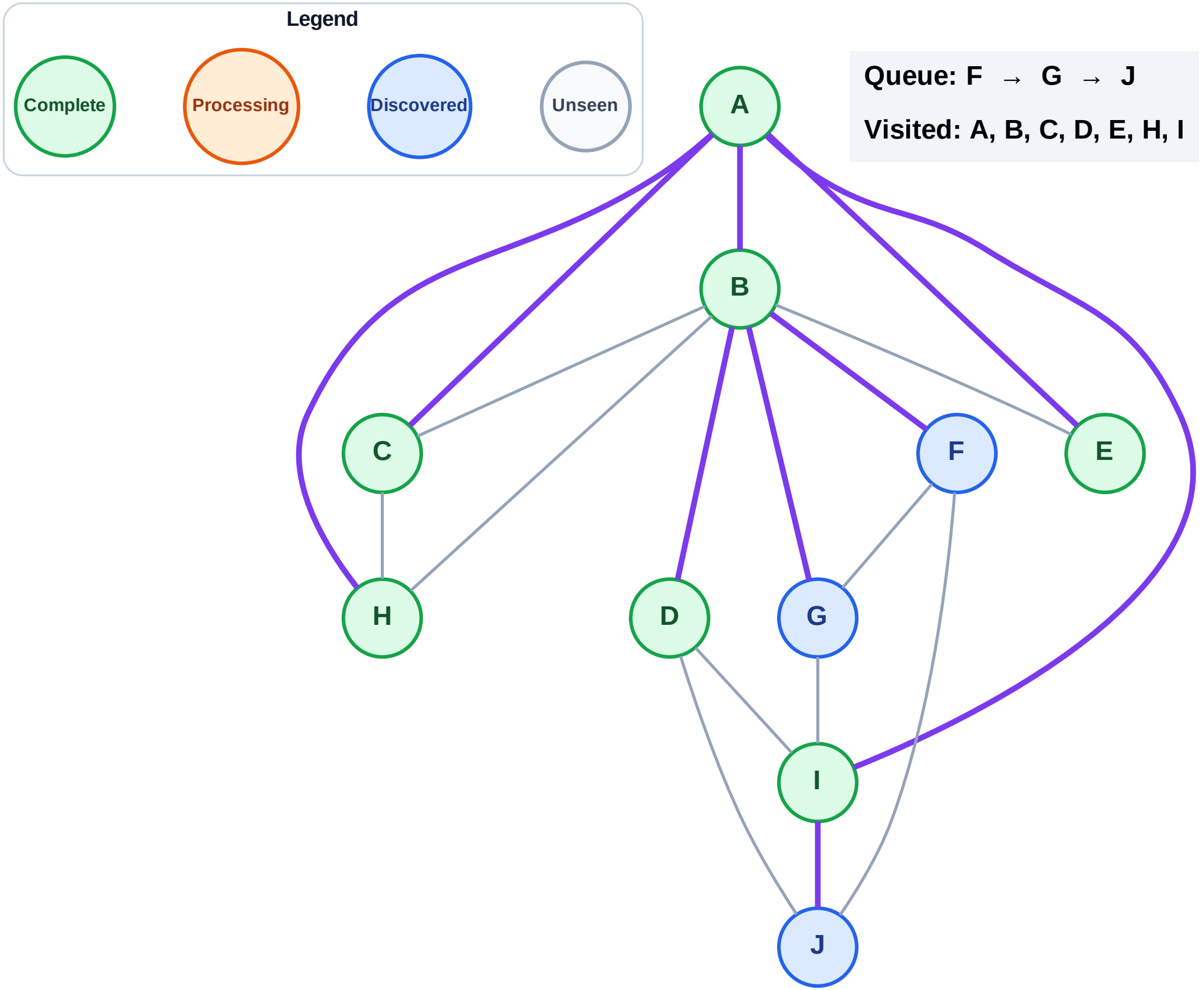
BFS Traversal

Step 14: Process node D



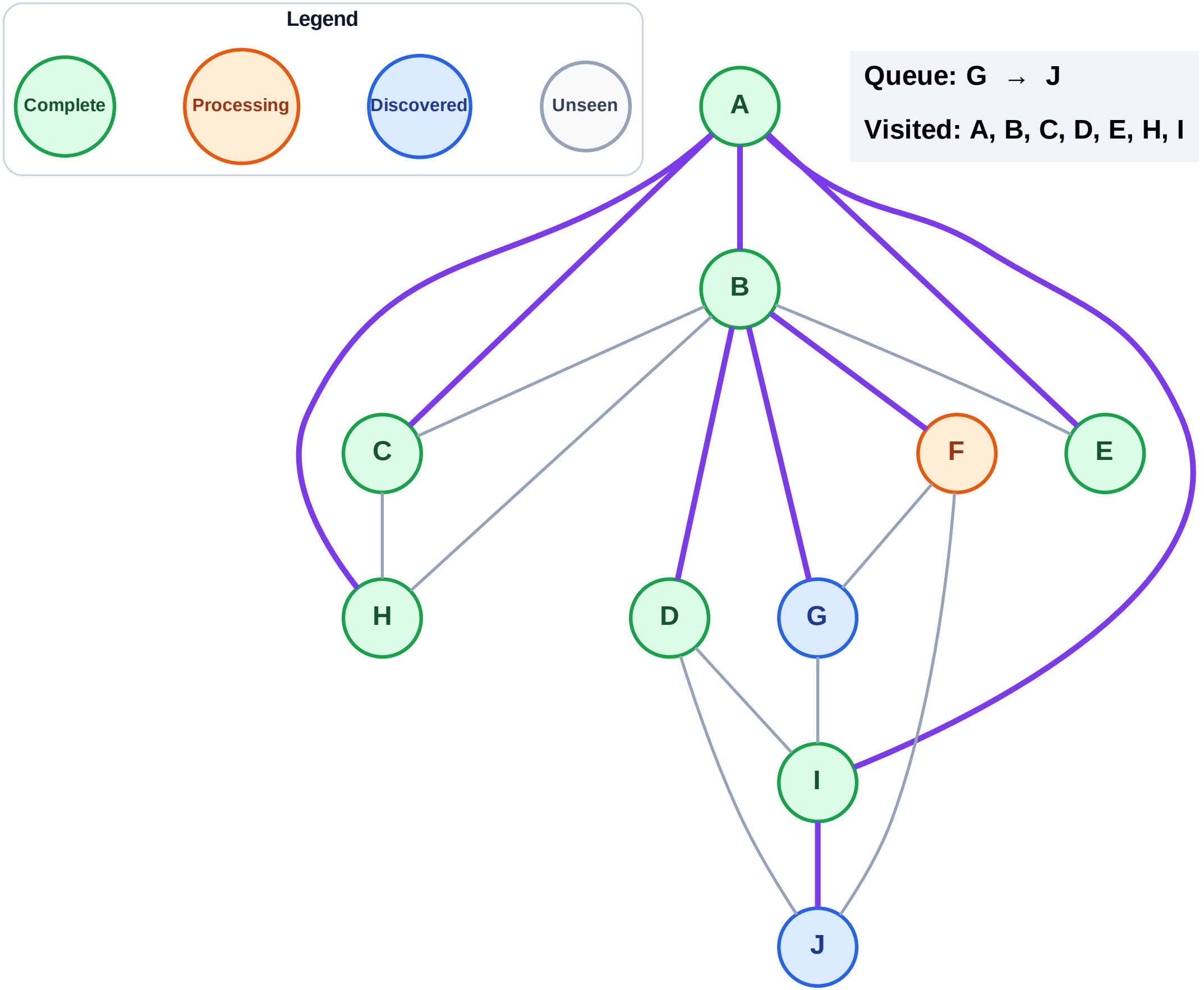
BFS Traversal

Step 15: No new neighbors from D



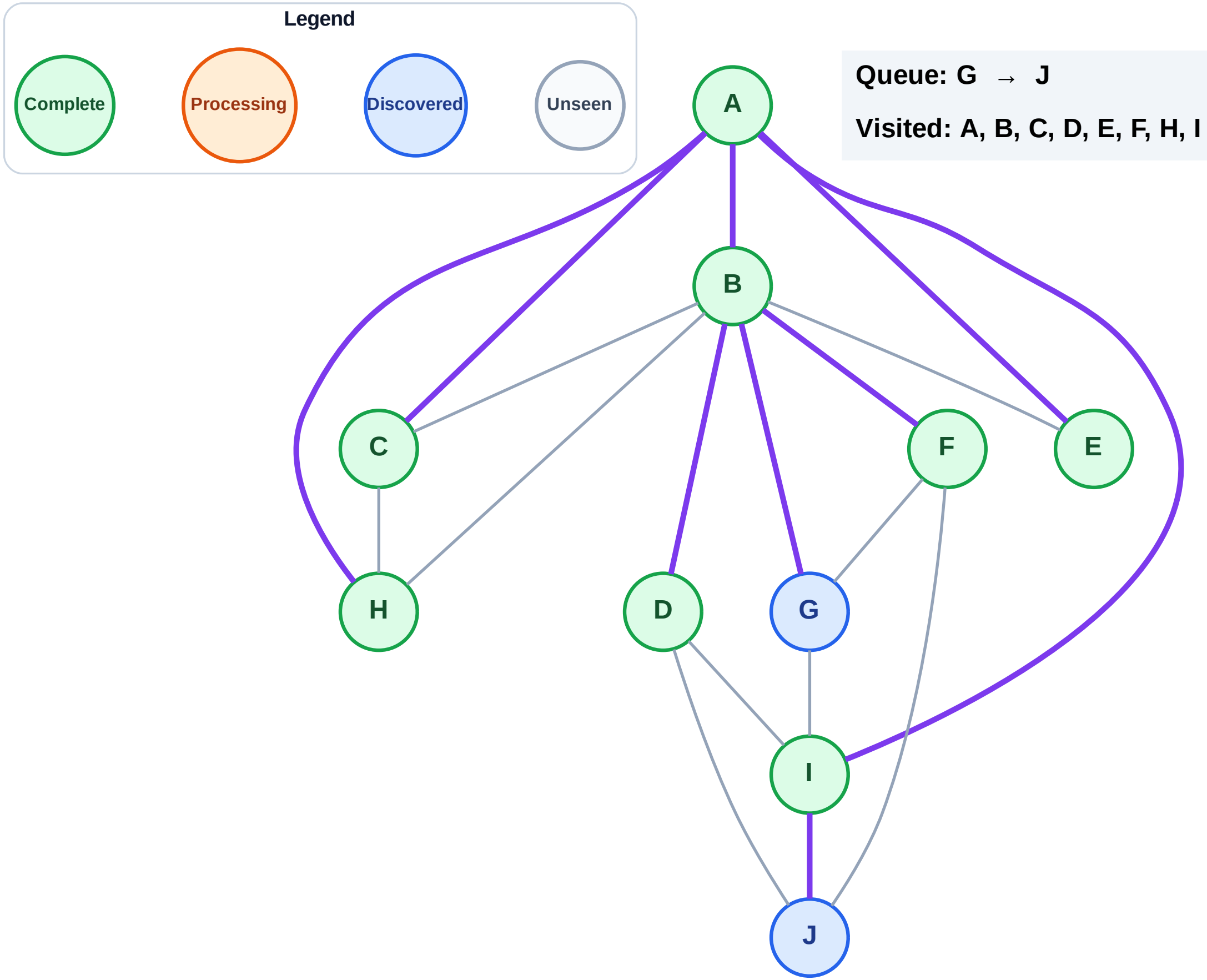
BFS Traversal

Step 16: Process node F



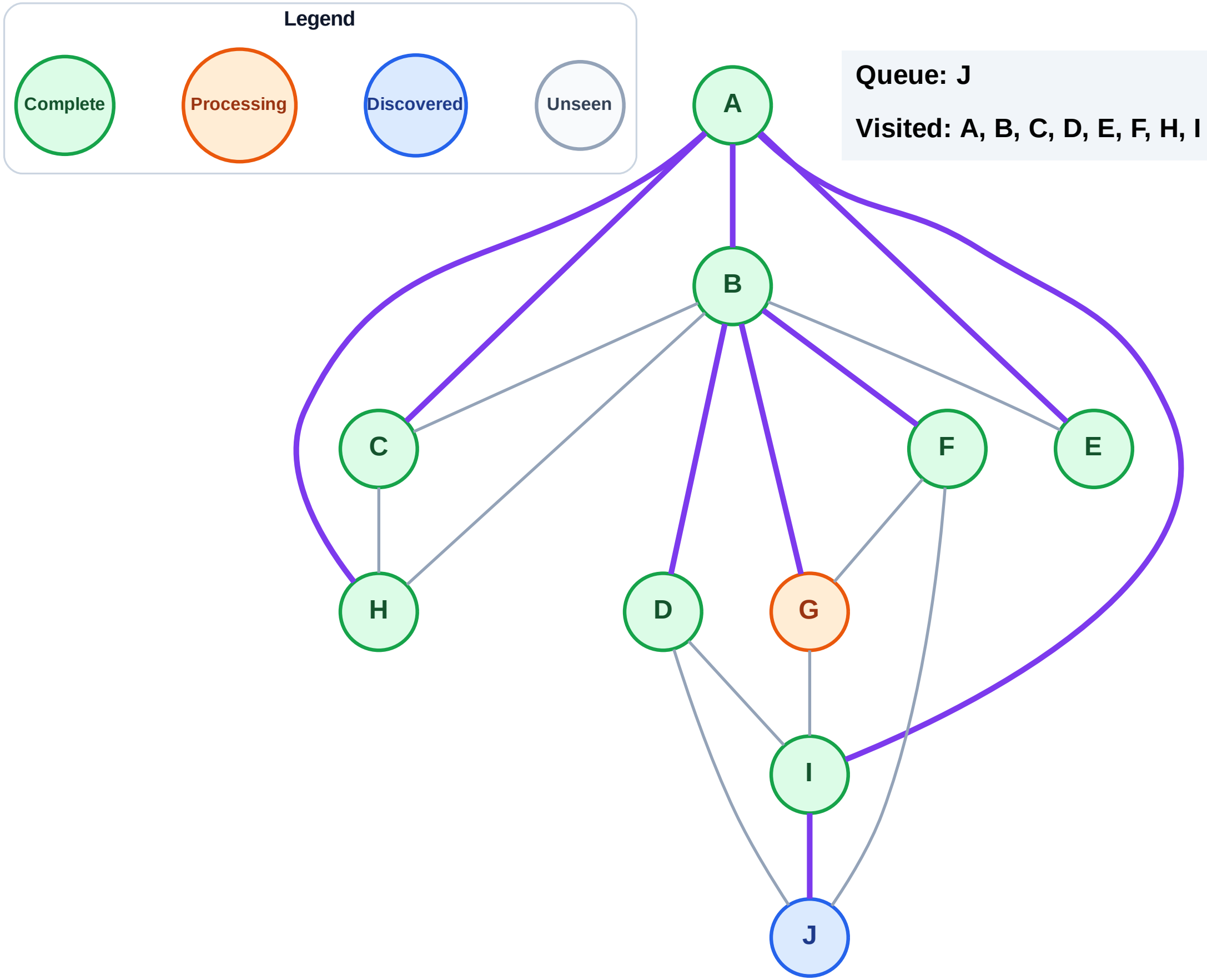
BFS Traversal

Step 17: No new neighbors from F



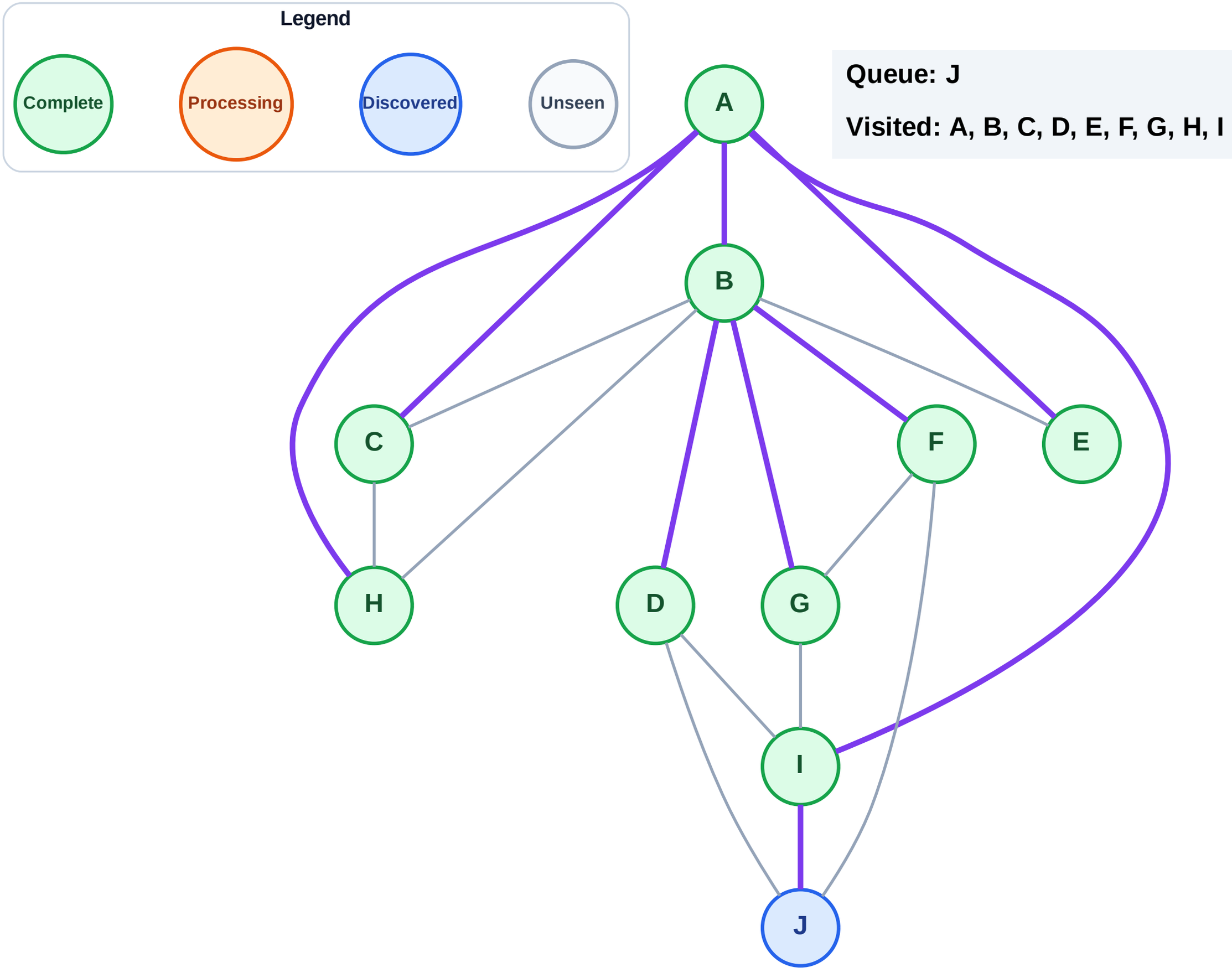
BFS Traversal

Step 18: Process node G



BFS Traversal

Step 19: No new neighbors from G

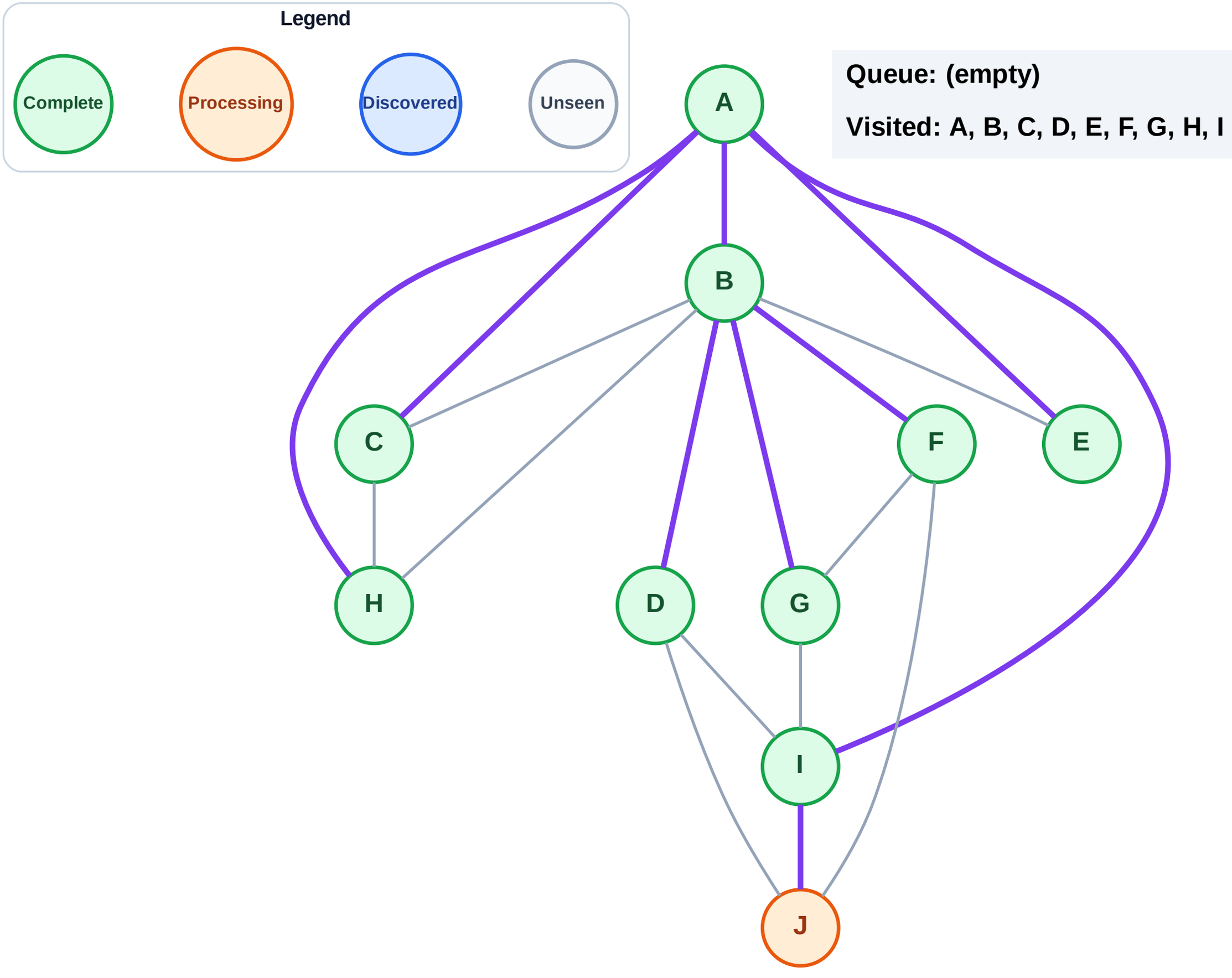


Queue: J

Visited: A, B, C, D, E, F, G, H, I

BFS Traversal

Step 20: Process node J

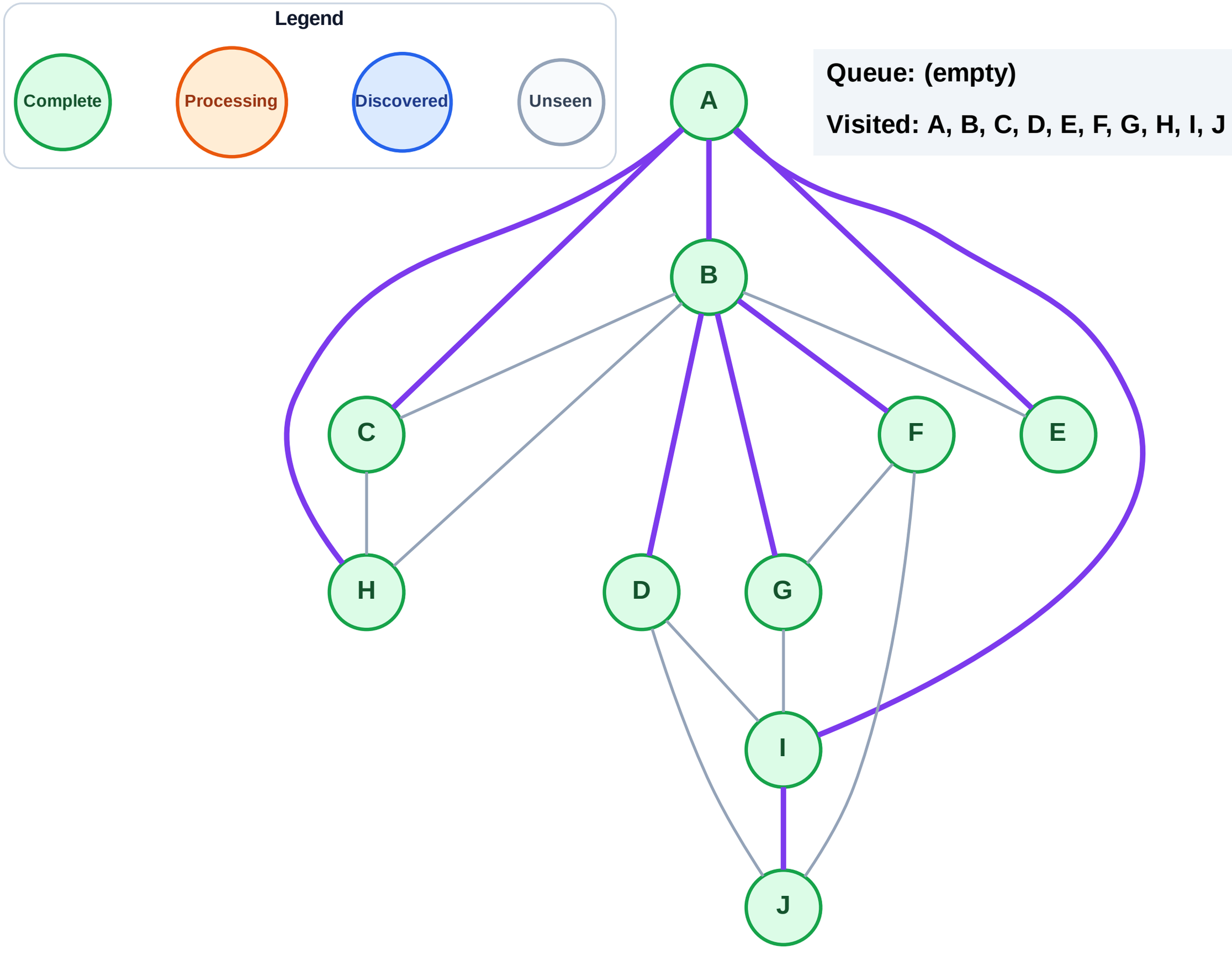


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I

BFS Traversal

Step 21: No new neighbors from J

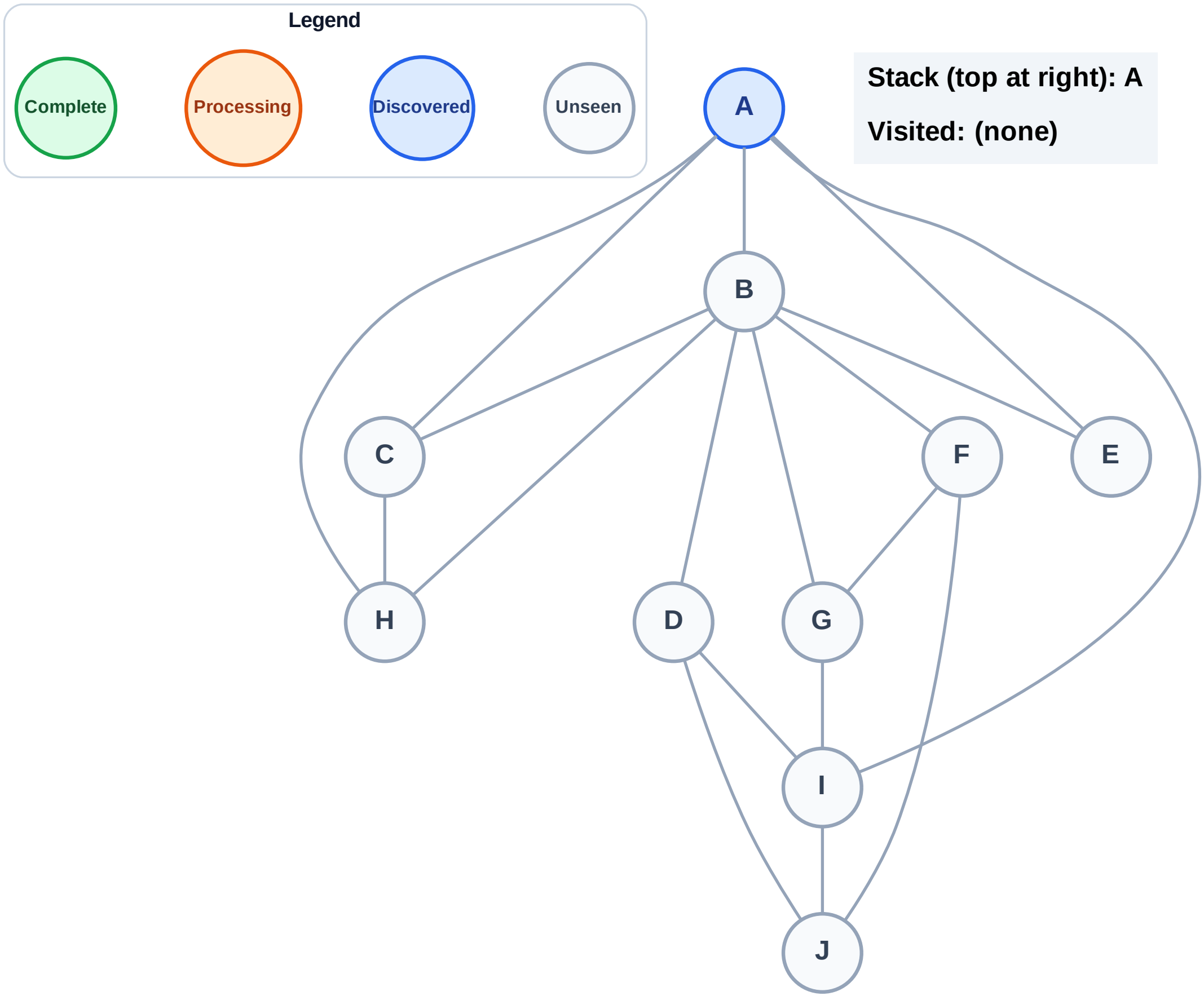


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

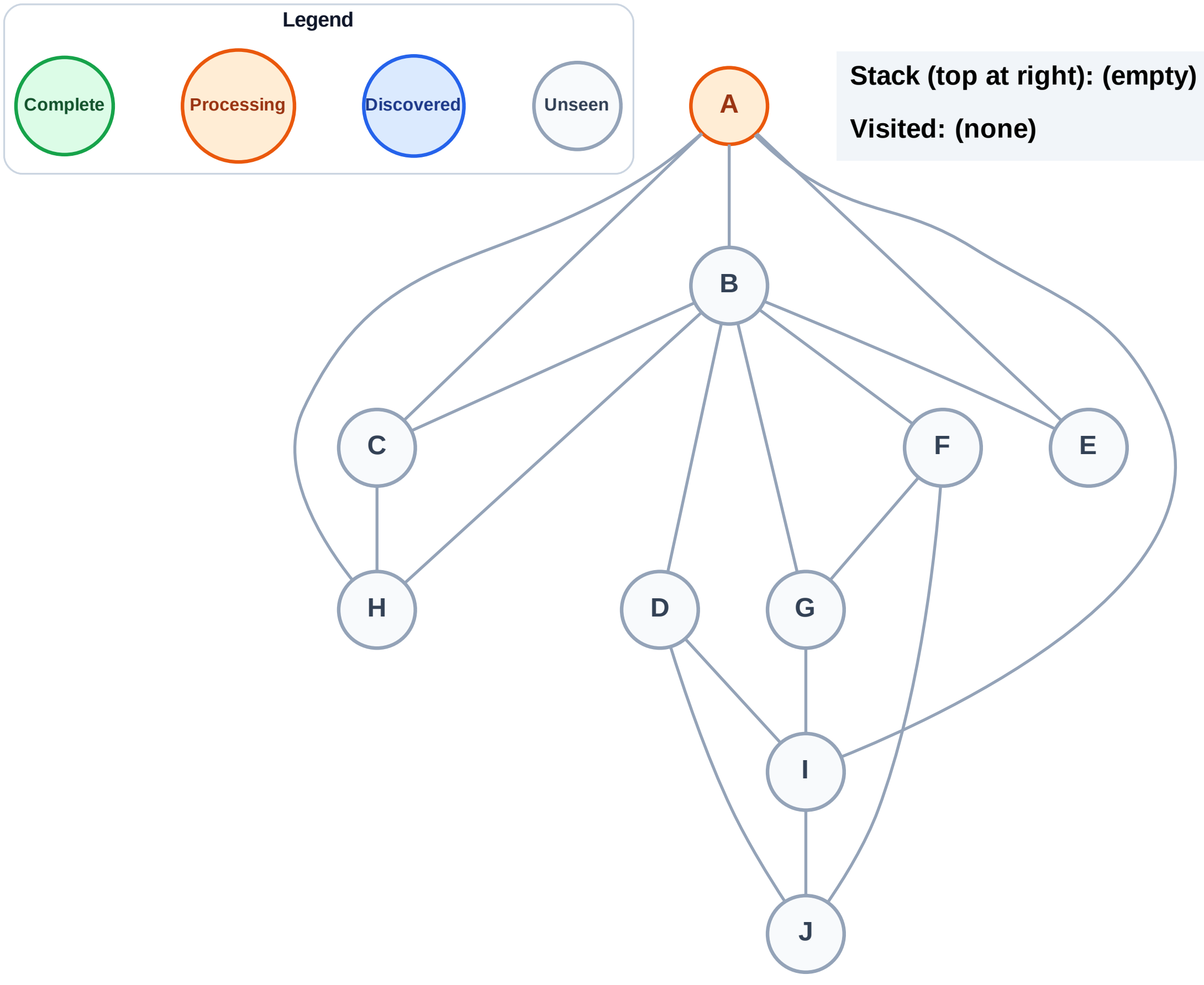
DFS Traversal

Step 1: Start at A



DFS Traversal

Step 2: Process node A

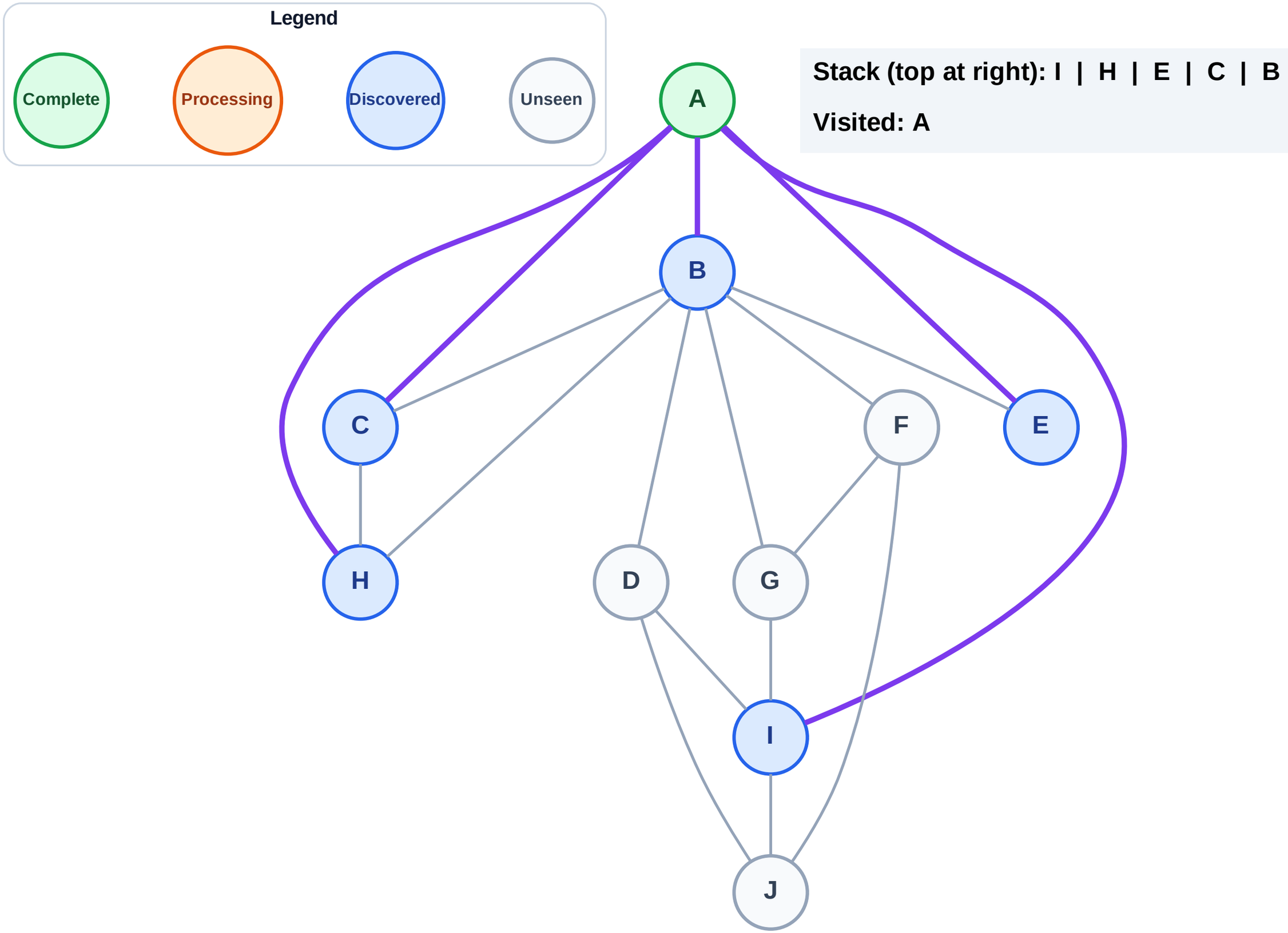


Stack (top at right): (empty)

Visited: (none)

DFS Traversal

Step 3: Discover I, H, E, C, B from A

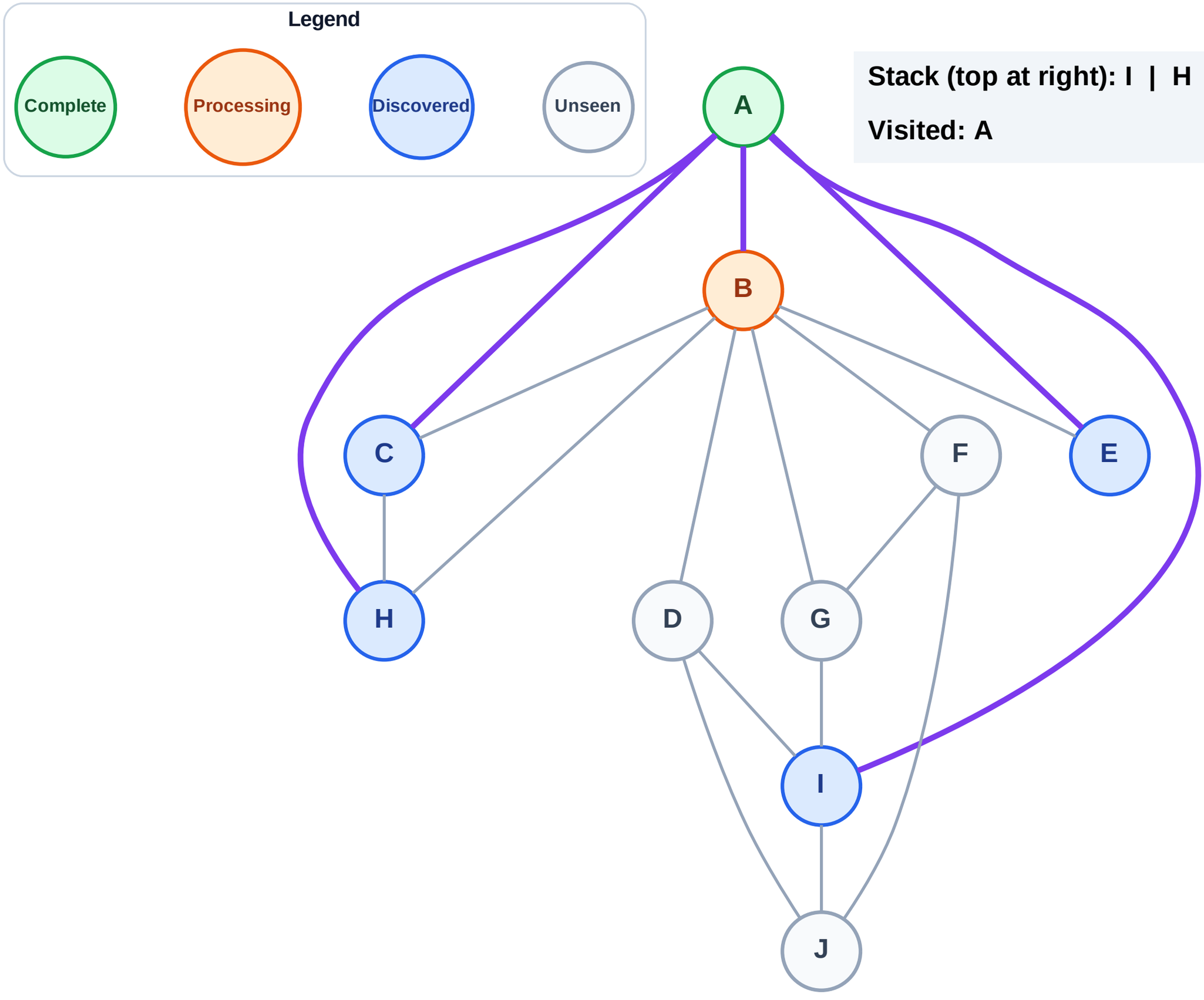


Stack (top at right): I | H | E | C | B

Visited: A

DFS Traversal

Step 4: Process node B

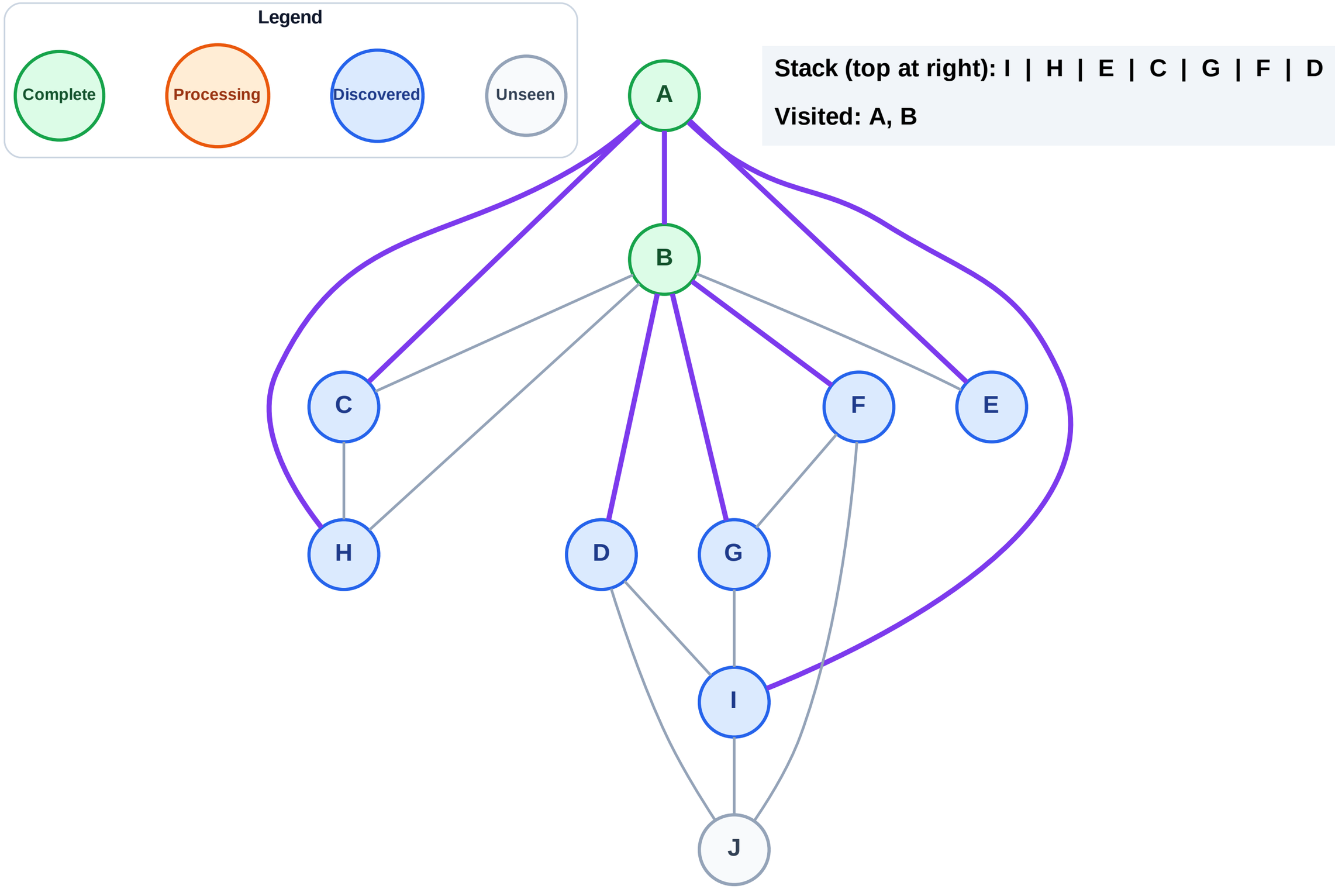


Stack (top at right): I | H | E | C

Visited: A

DFS Traversal

Step 5: Discover G, F, D from B

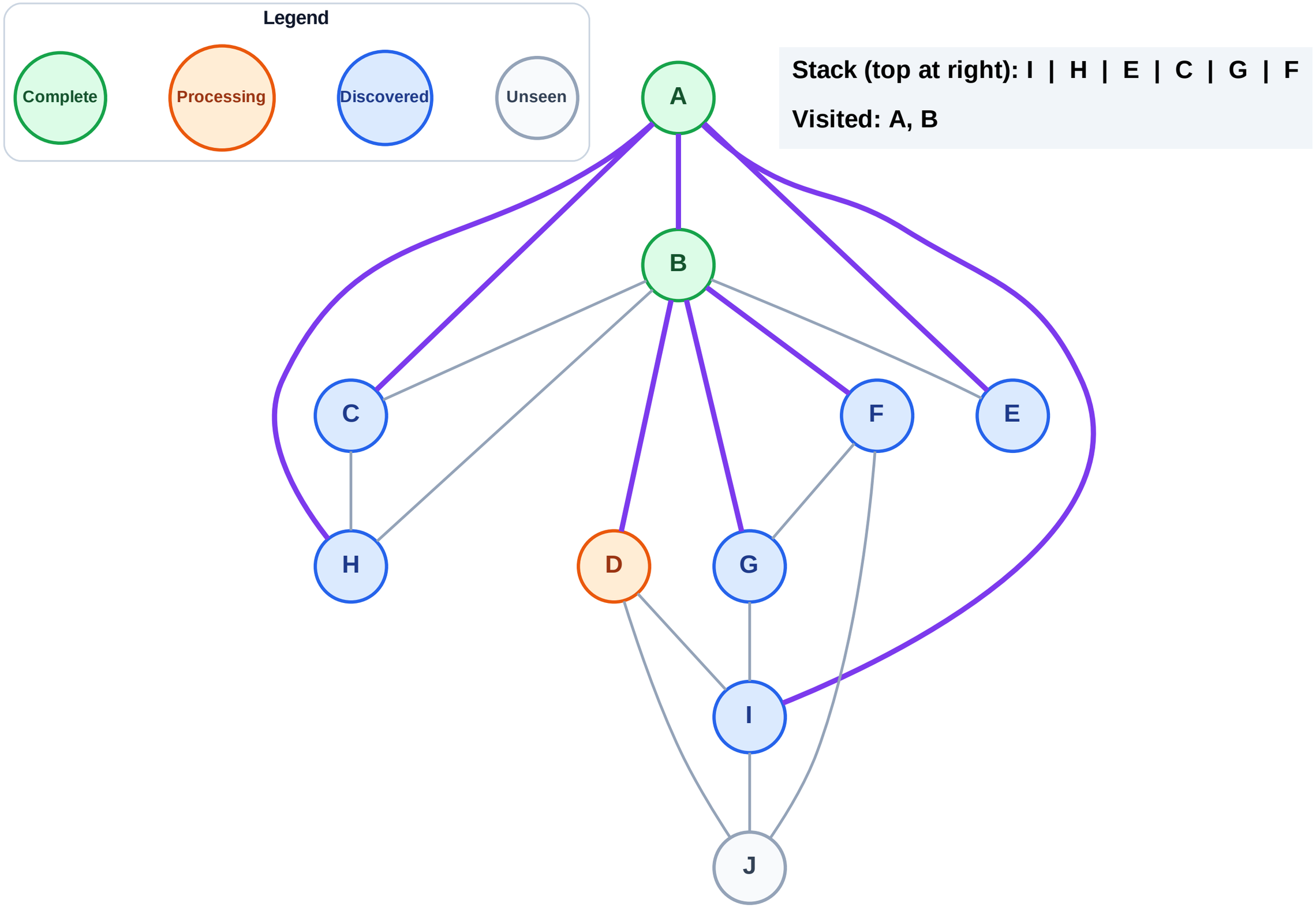


Stack (top at right): I | H | E | C | G | F | D

Visited: A, B

DFS Traversal

Step 6: Process node D

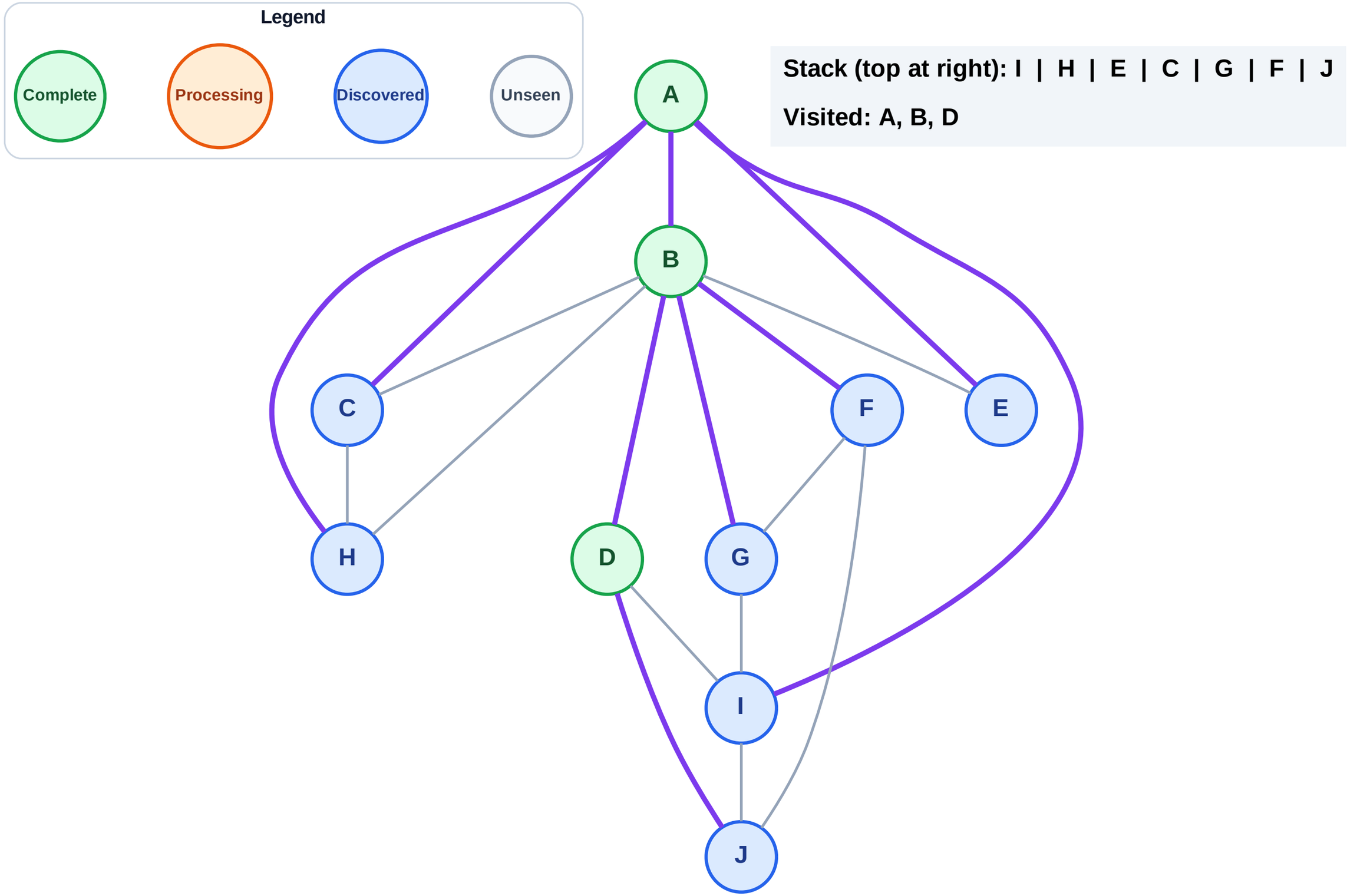


Stack (top at right): I | H | E | C | G | F

Visited: A, B

DFS Traversal

Step 7: Discover J from D

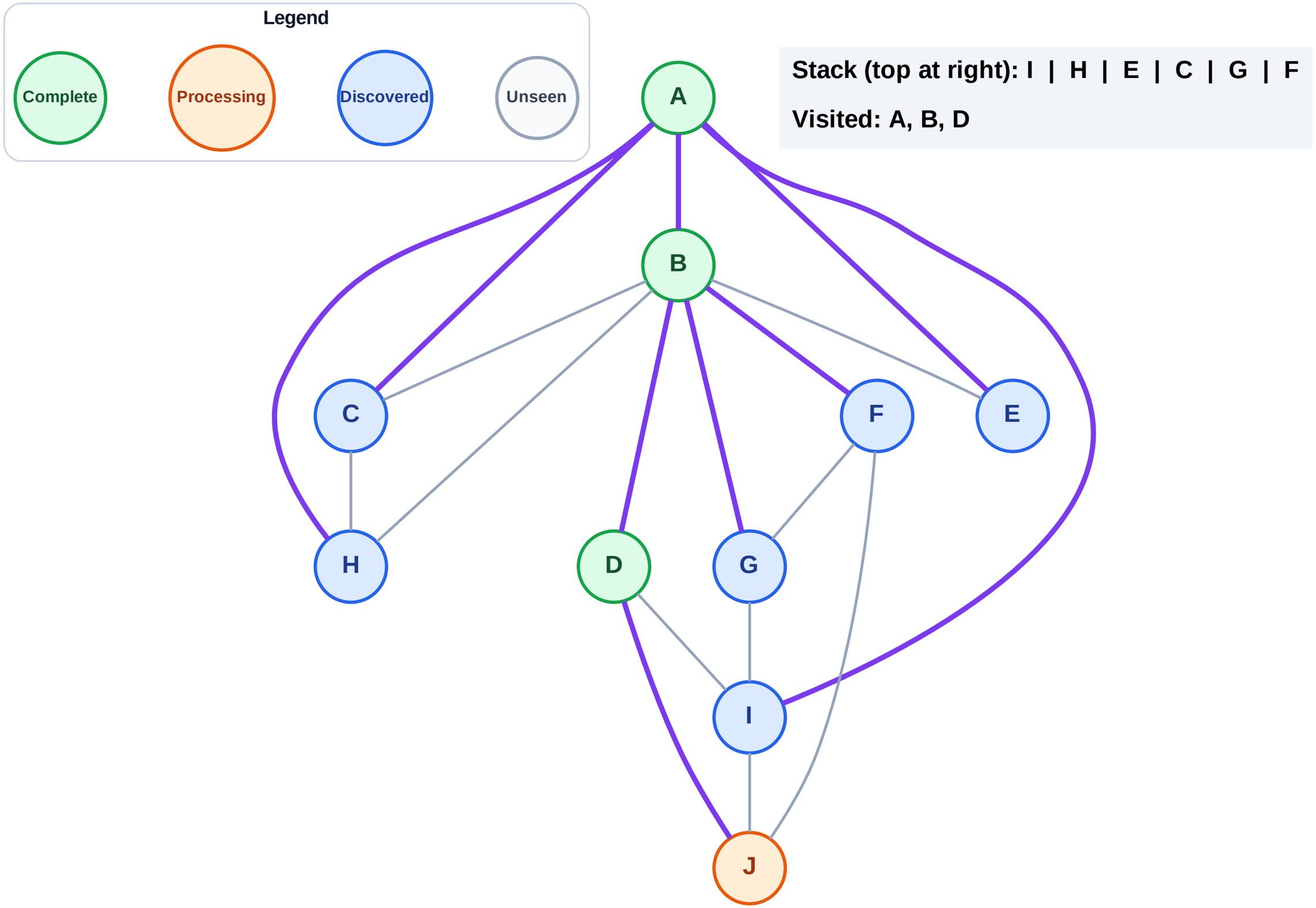


Stack (top at right): I | H | E | C | G | F | J

Visited: A, B, D

DFS Traversal

Step 8: Process node J

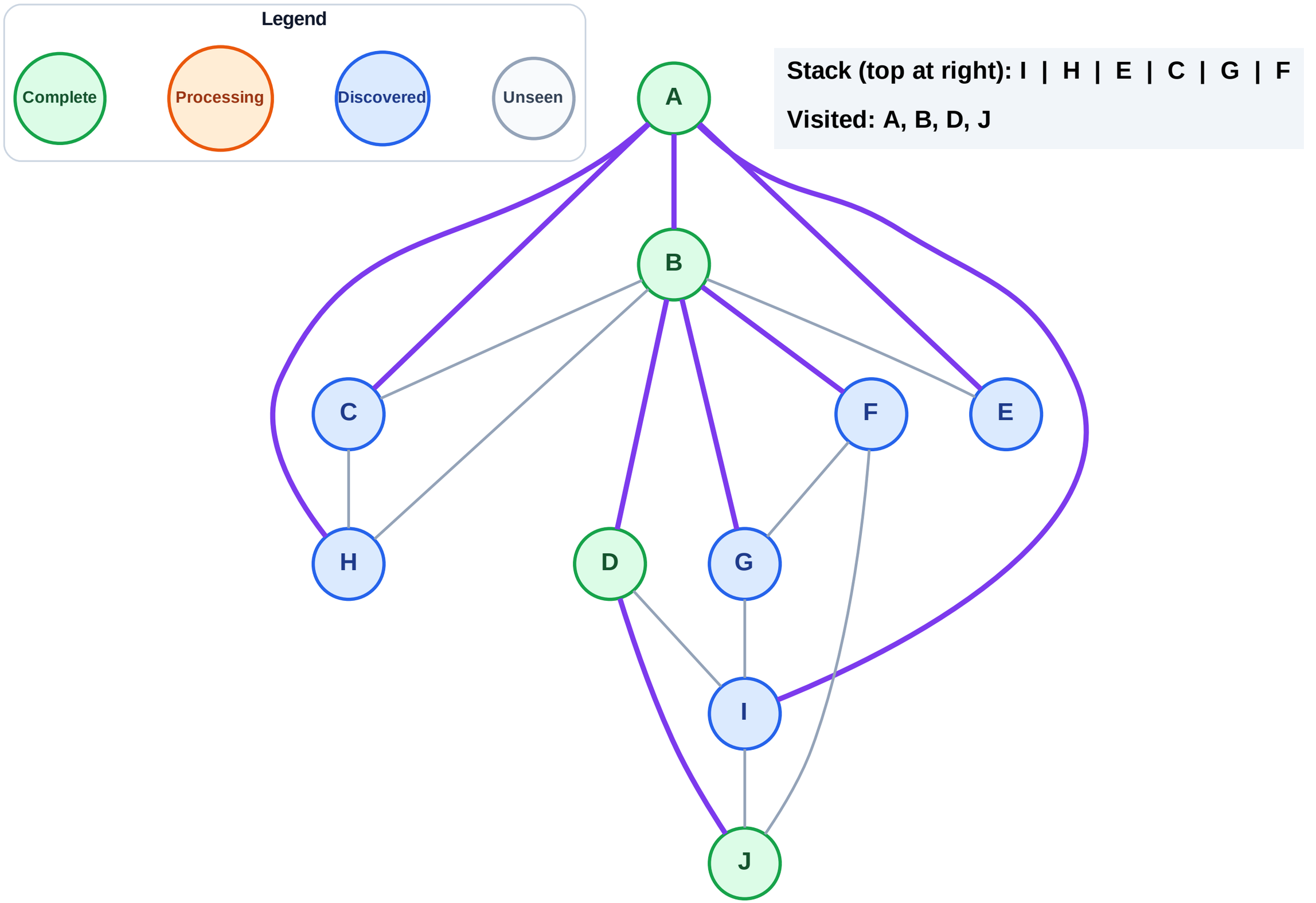


Stack (top at right): I | H | E | C | G | F

Visited: A, B, D

DFS Traversal

Step 9: No new neighbors from J

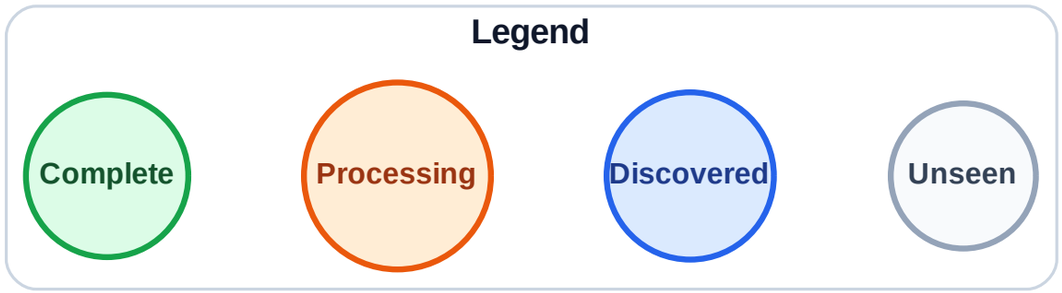


Stack (top at right): I | H | E | C | G | F

Visited: A, B, D, J

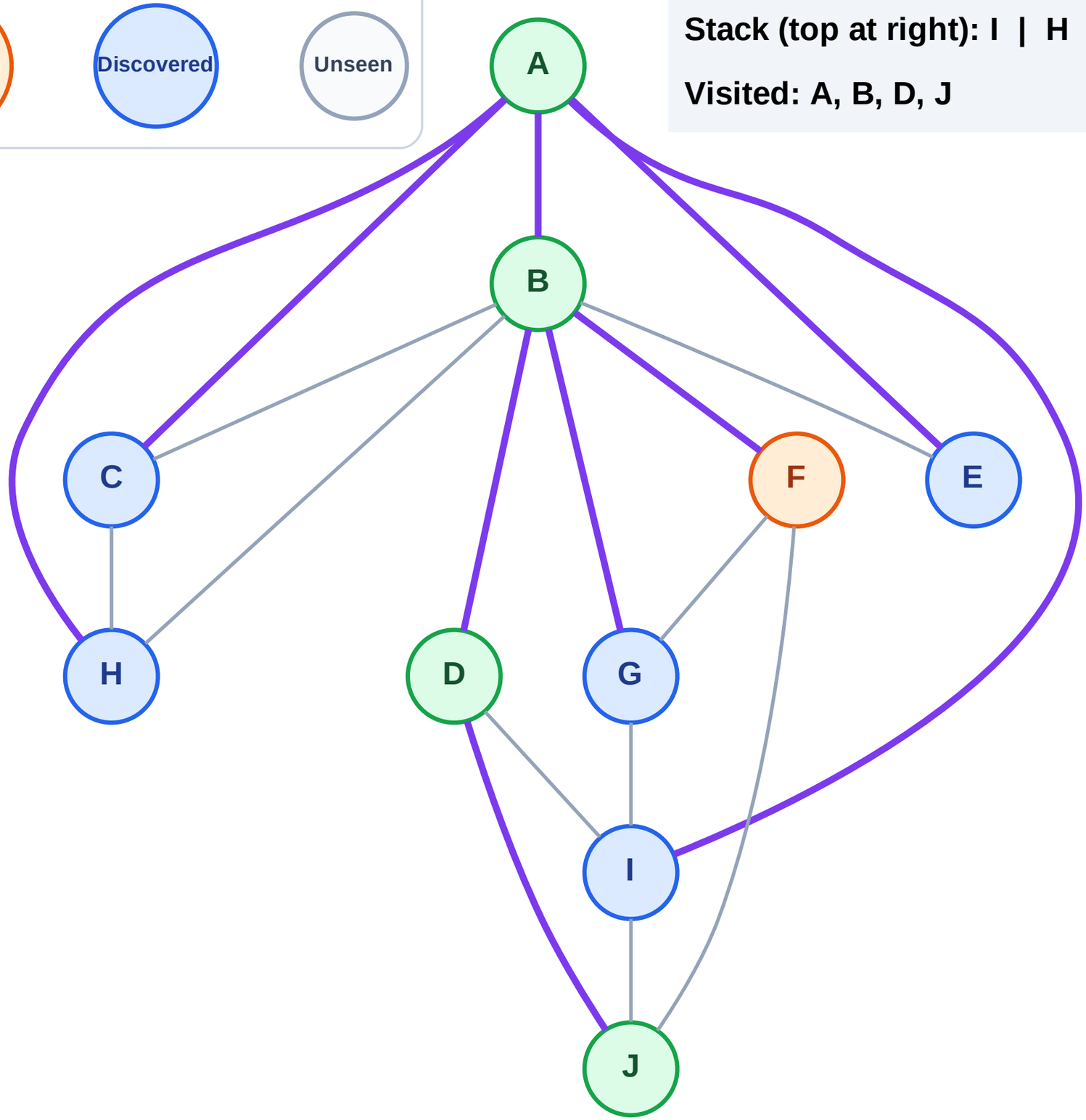
DFS Traversal

Step 10: Process node F



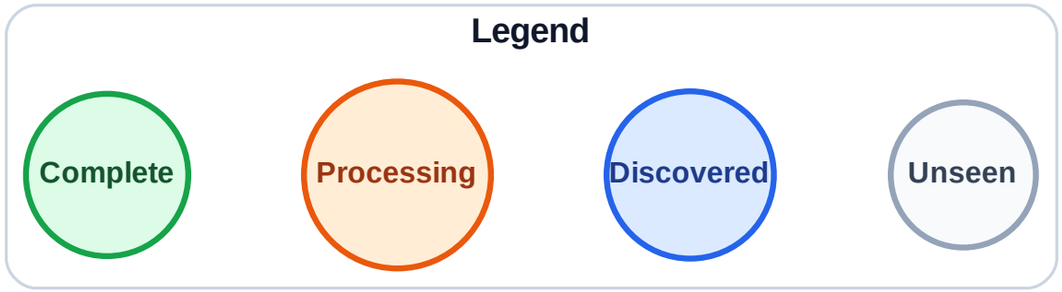
Stack (top at right): I | H | E | C | G

Visited: A, B, D, J



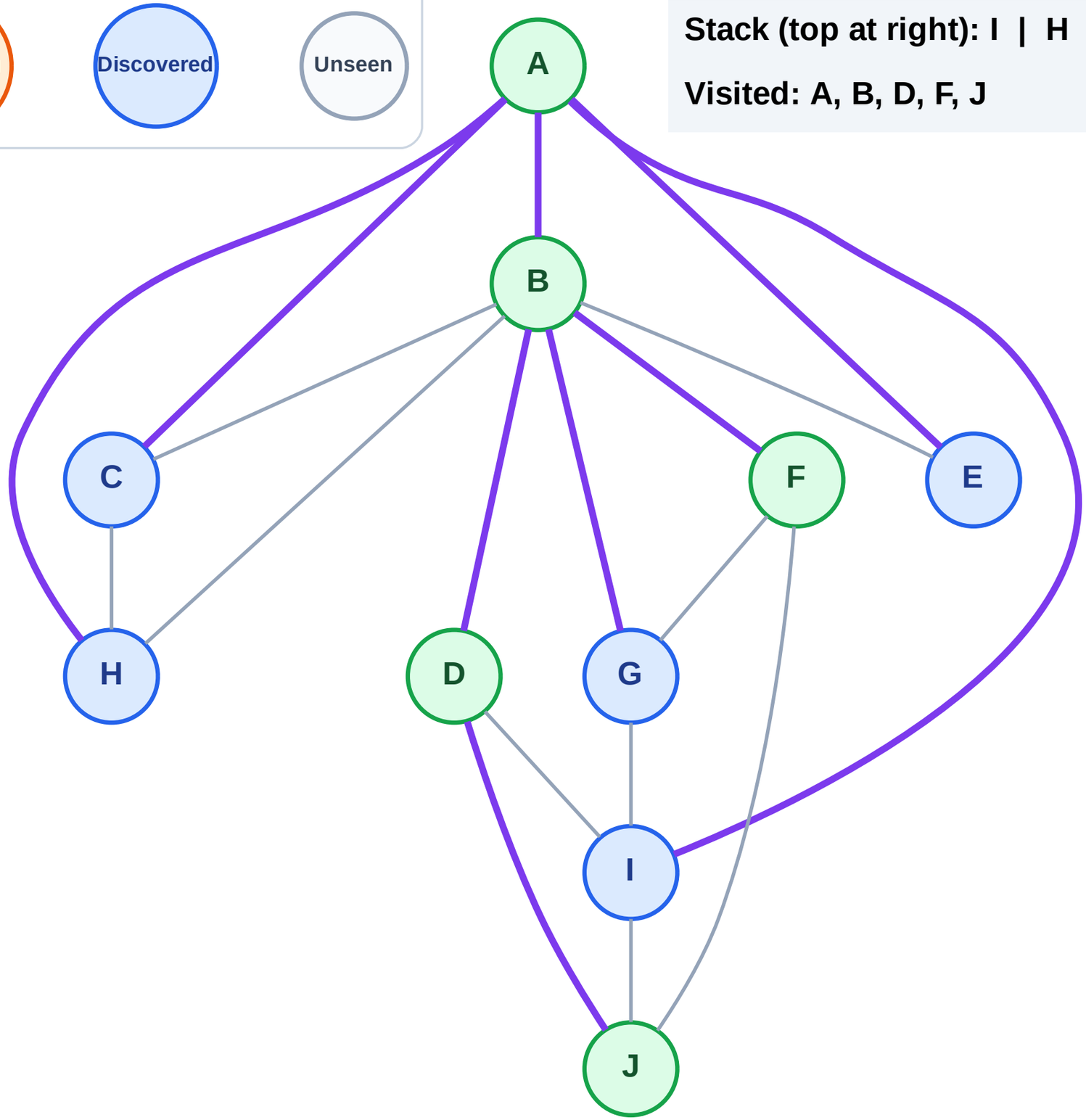
DFS Traversal

Step 11: No new neighbors from F



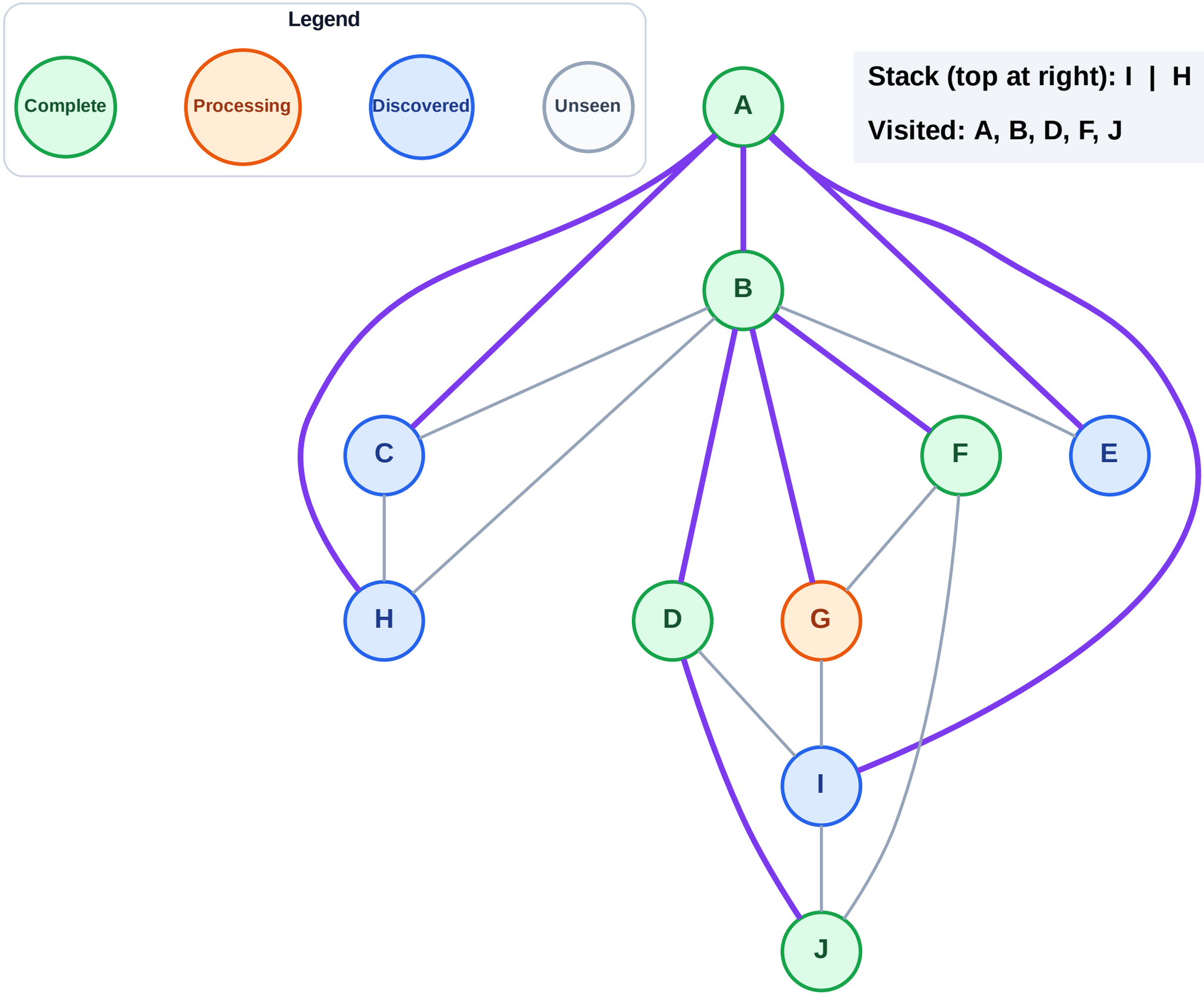
Stack (top at right): I | H | E | C | G

Visited: A, B, D, F, J



DFS Traversal

Step 12: Process node G

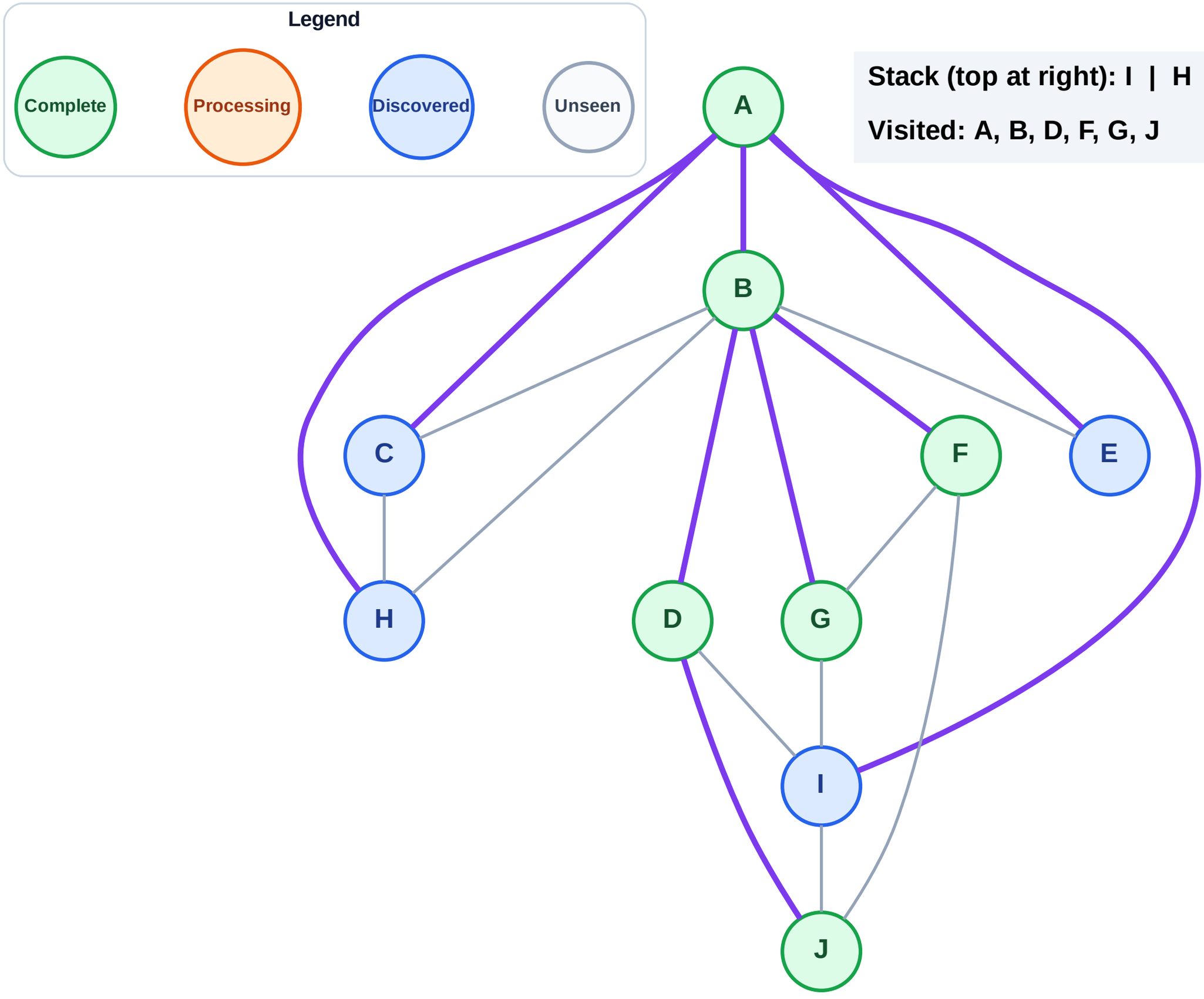


Stack (top at right): I | H | E | C

Visited: A, B, D, F, J

DFS Traversal

Step 13: No new neighbors from G

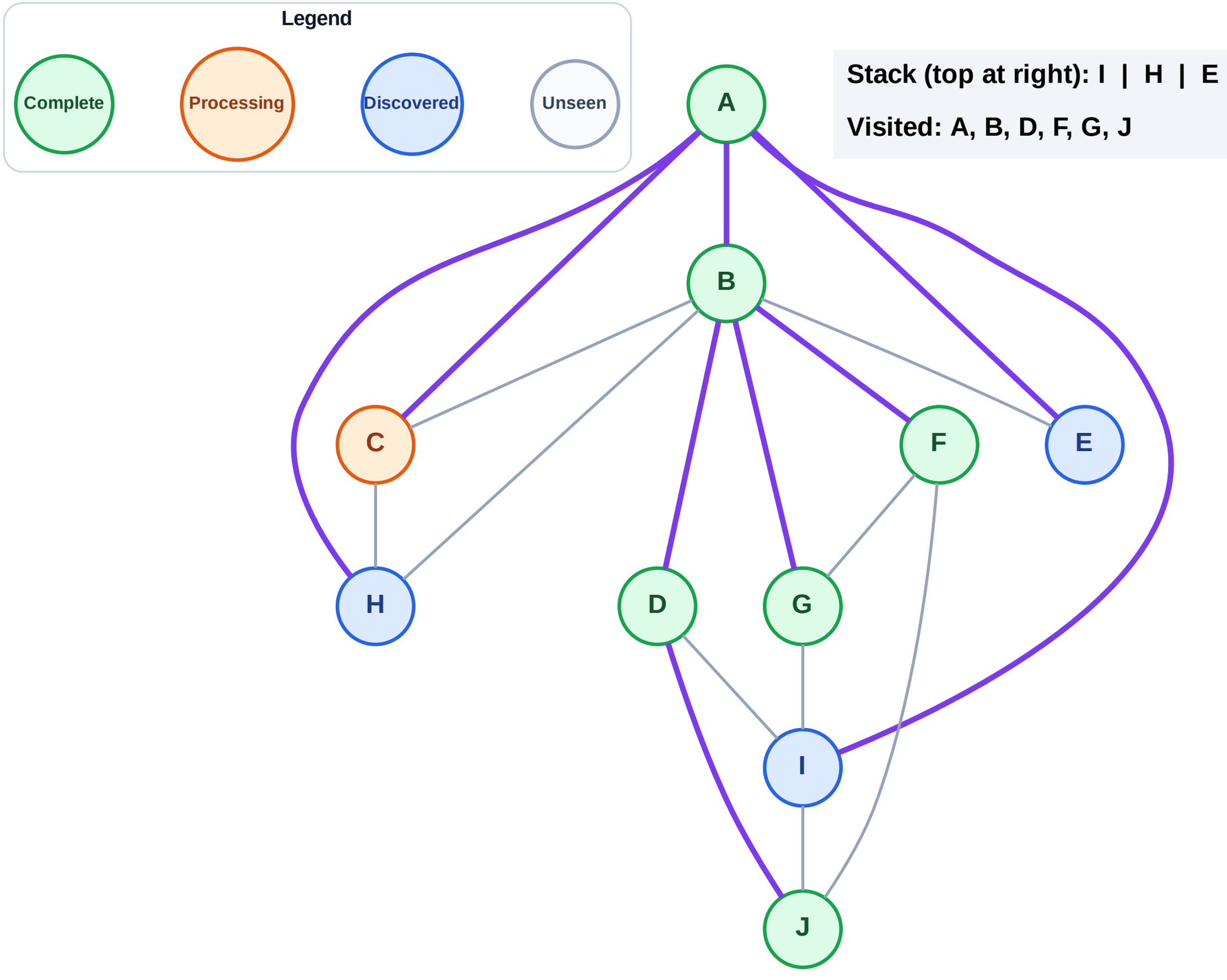


Stack (top at right): I | H | E | C

Visited: A, B, D, F, G, J

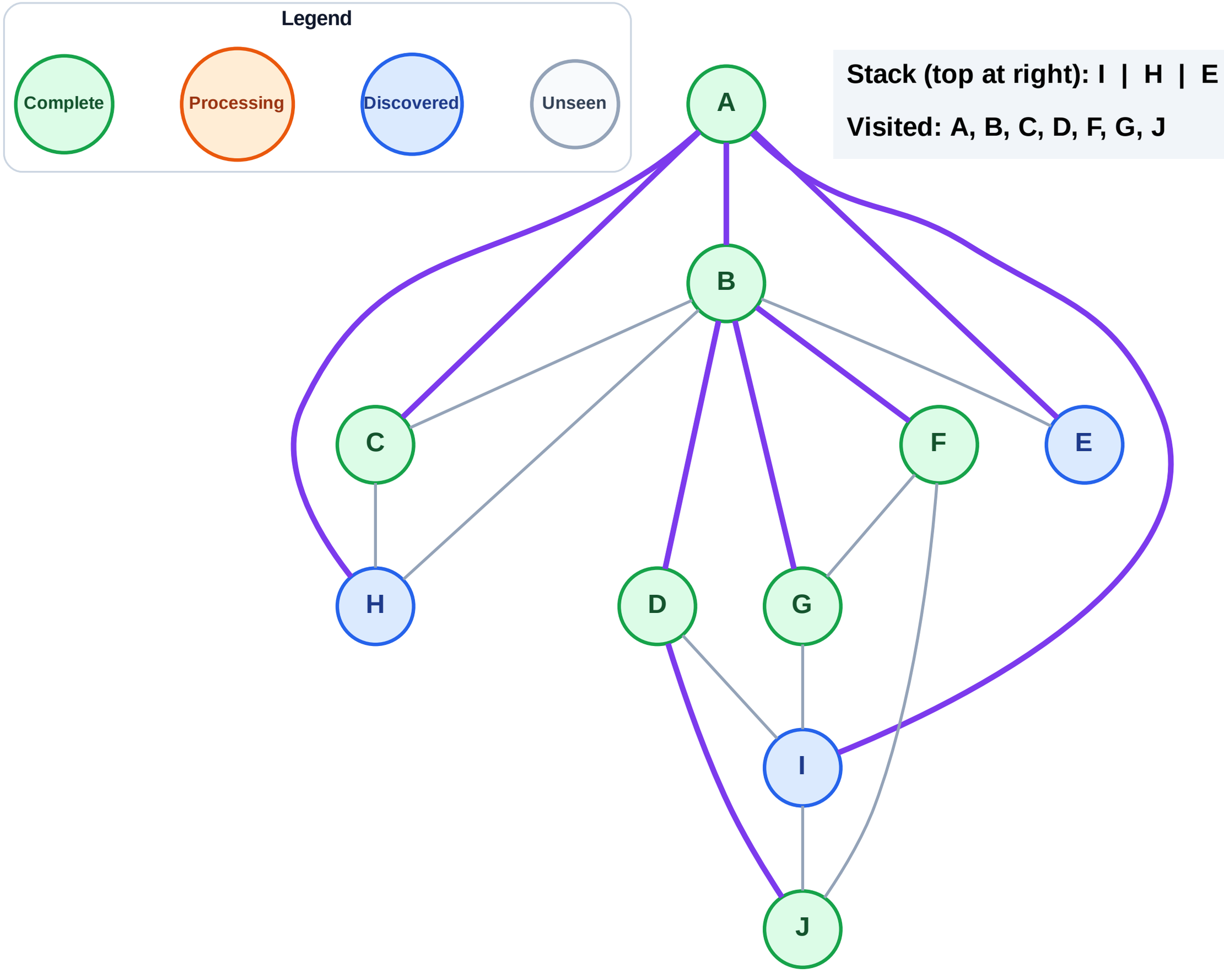
DFS Traversal

Step 14: Process node C



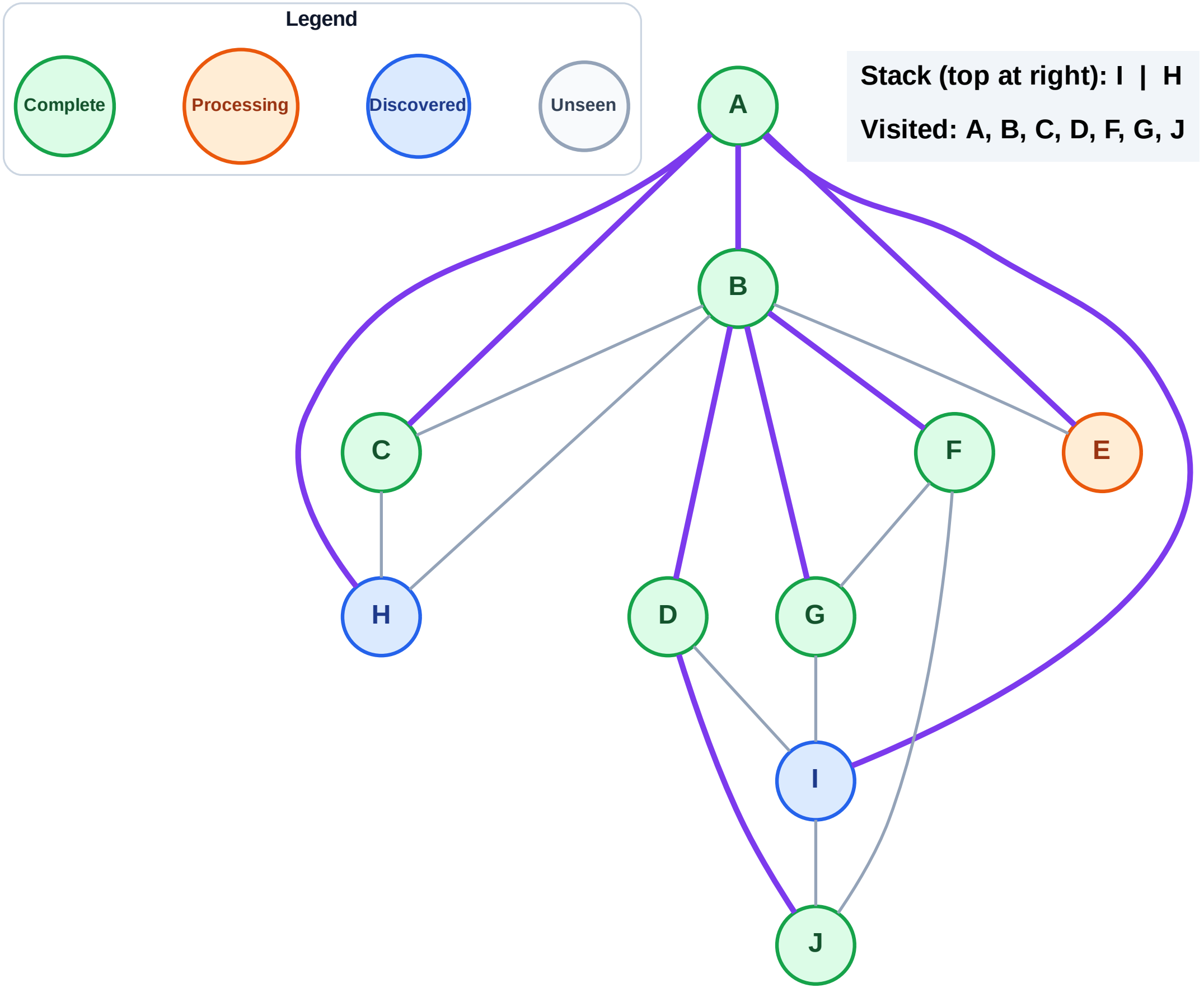
DFS Traversal

Step 15: No new neighbors from C



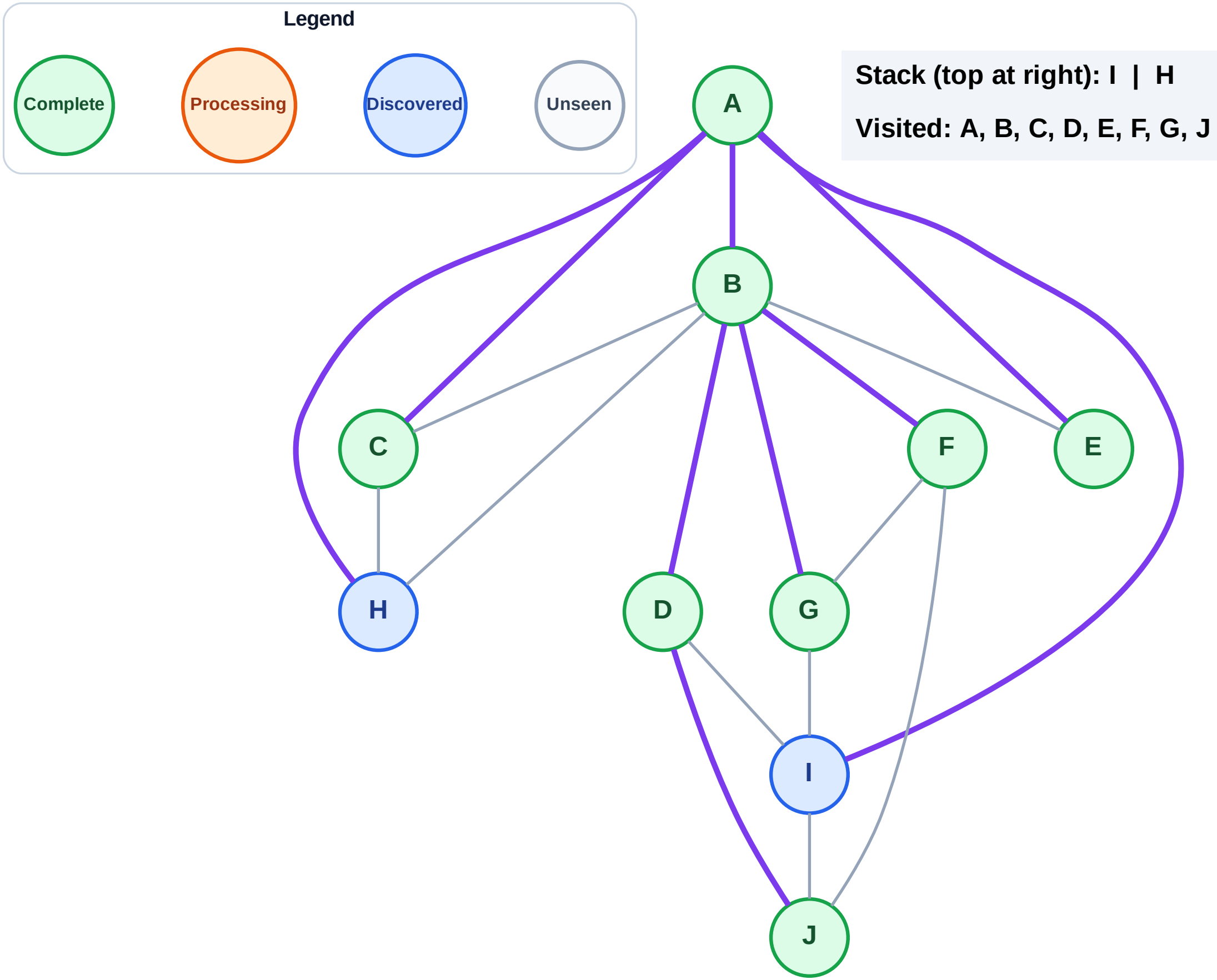
DFS Traversal

Step 16: Process node E



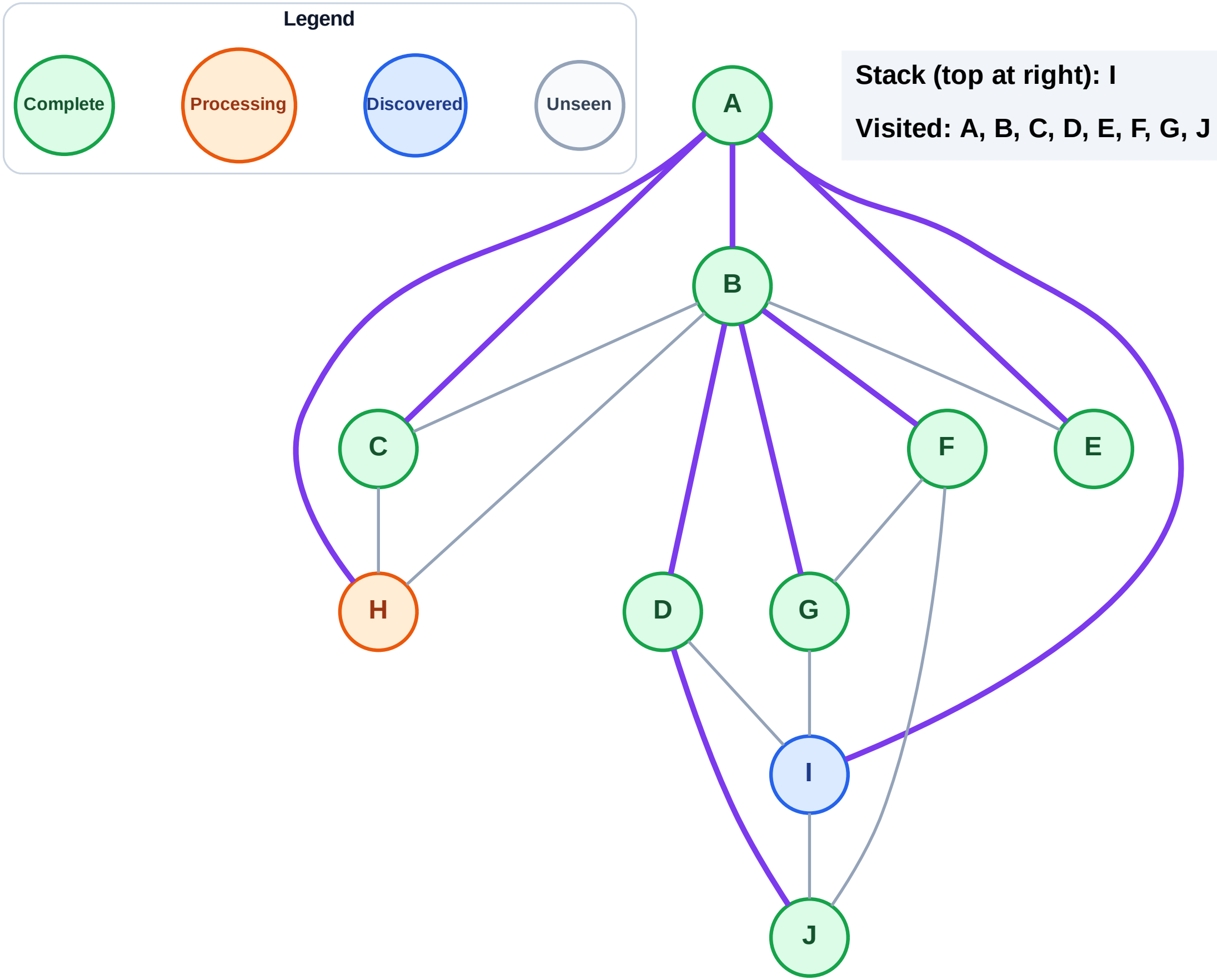
DFS Traversal

Step 17: No new neighbors from E



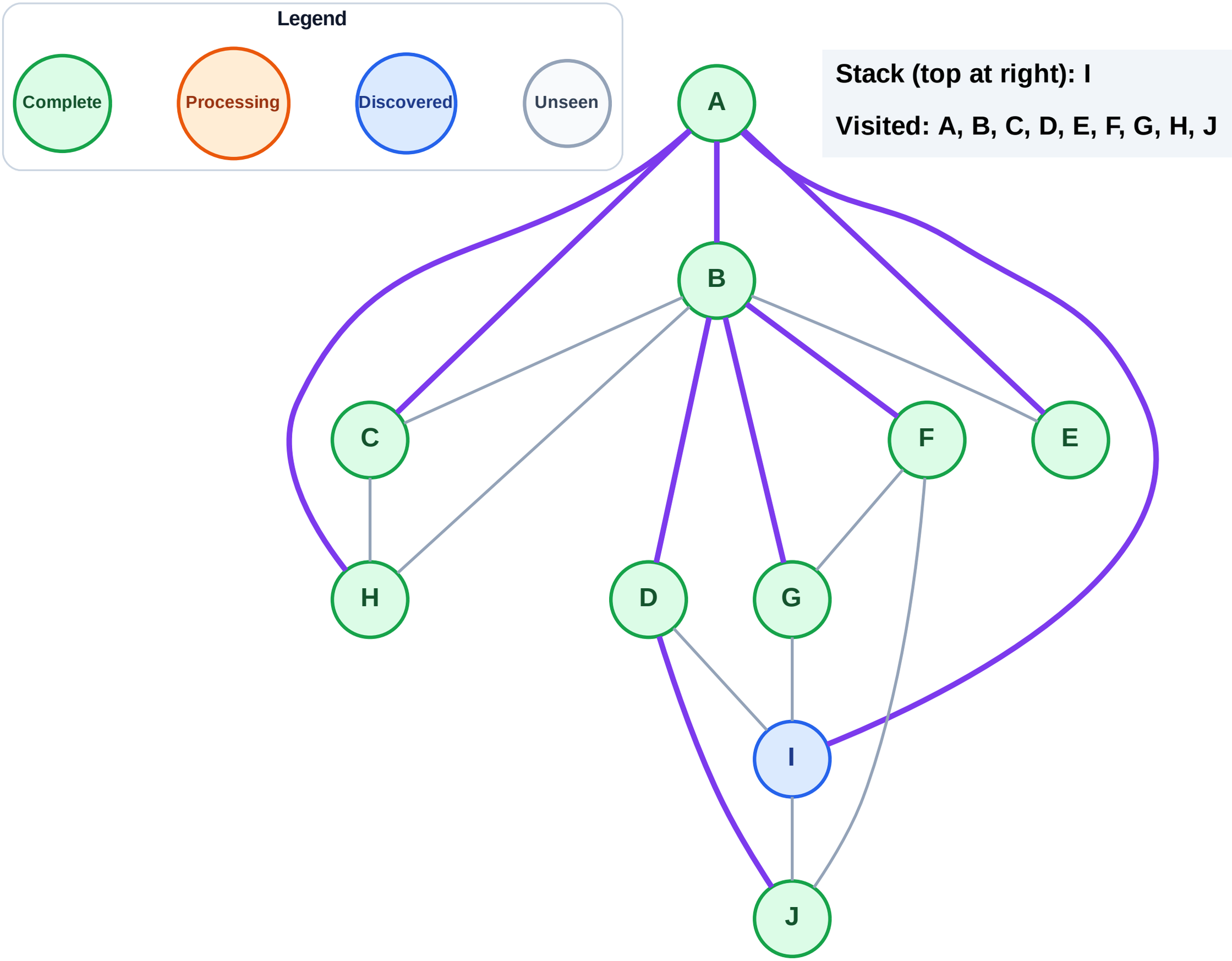
DFS Traversal

Step 18: Process node H



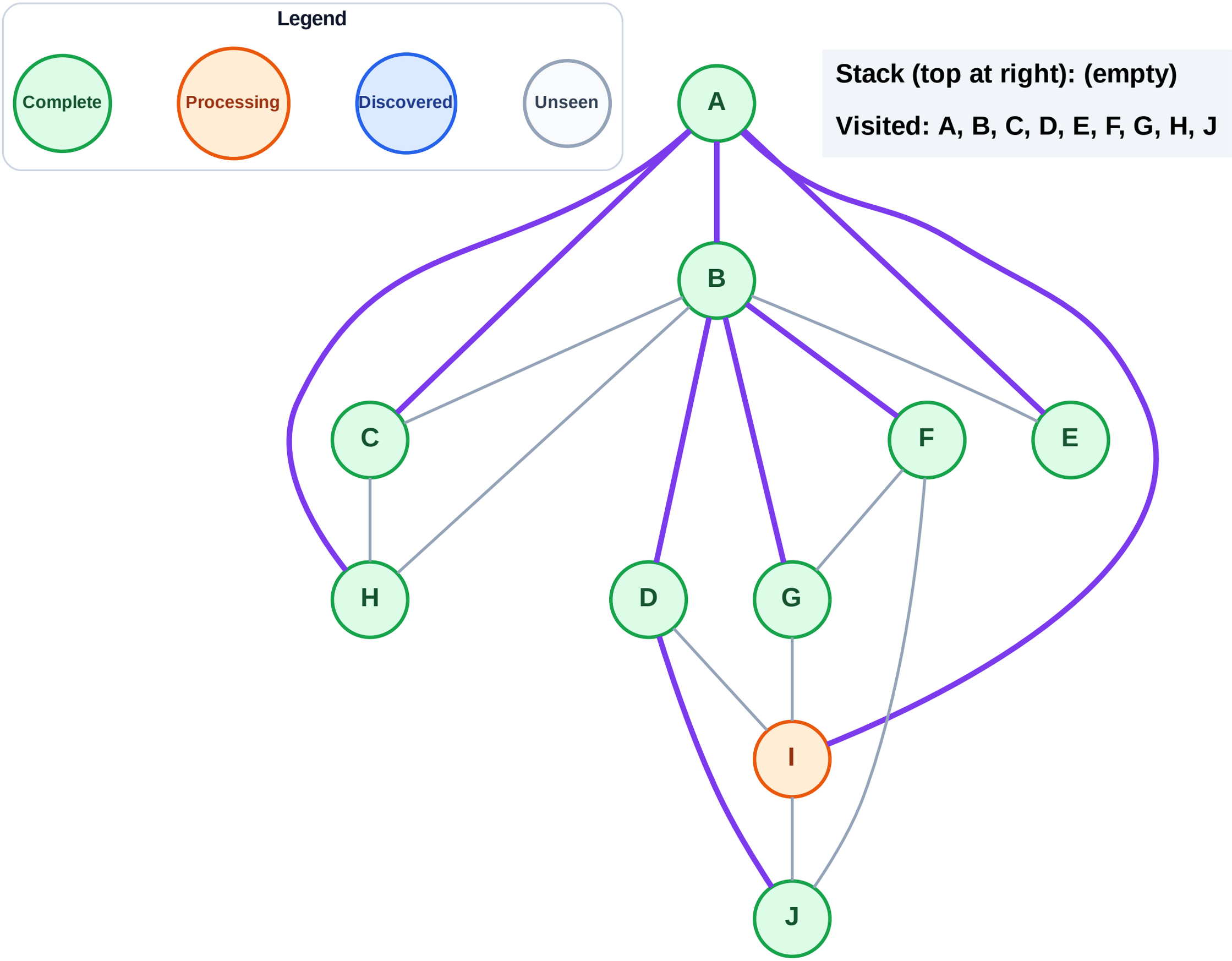
DFS Traversal

Step 19: No new neighbors from H



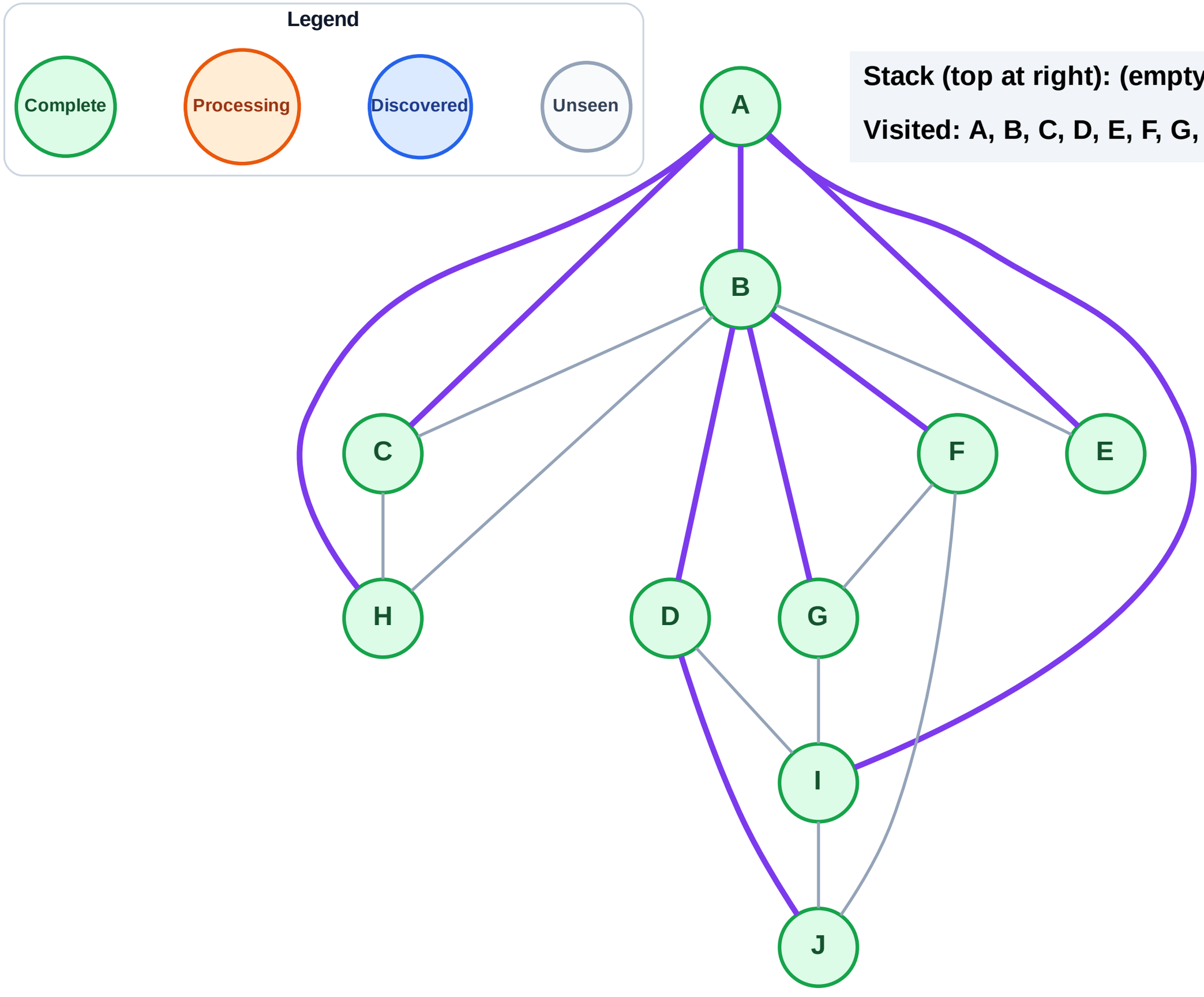
DFS Traversal

Step 20: Process node I



DFS Traversal

Step 21: No new neighbors from I



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J